01 **URBAN SHELTER**
Urban complex

02 **INVASION BY SLIME MOULD**
Renewal of Abandoned metro space

03 **NAIROBI CYCLE**
Slum rehabilitation

04 **URBAN HABITAT**
Rehabilitation of industrial heritage

05 **SPORTZENTRUM**
Design for Swimming Pool Cover Steel Structure Space Frame Steel Truss

URBAN DEISIGN PORTFOLIO

HANYUN HU

Selected works from 2020-2024

13752296518@163.com

Application number:23106433
Application to UCL
For MArch Urban Design

No.1

URBAN SHELTER

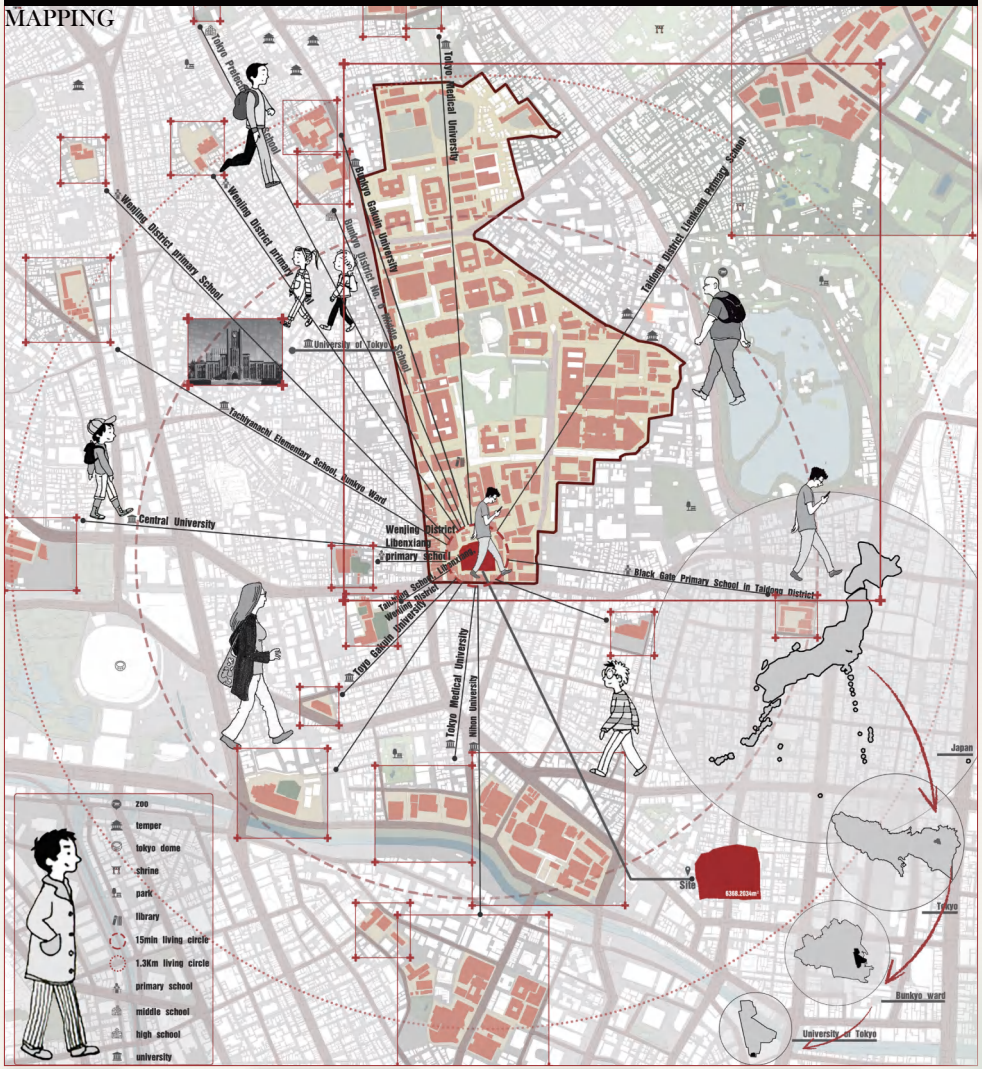
1 penny

Location:University of Tokyo
Academic individual work
Area:10,050m²
Duration:2023.9-12

URBAN SHELTER to be completed by January 2030.Statistics show that the incidence of depression in Japan is as high as 17.7%, the suicide rate is as high as 17.5%, &vicious incidents are frequent, so it's urgent to solve the political depressive tendencies.A team of Japanese researchers has published a report claiming that video games have the effect of improving mental health and enhancing life satisfaction. urbanshelter combines the worldview of the best game of the year - THE LEGEND OF ZELDA from Nintendo of Japan - with game mechanics such as TRAFFIC PATTERNS, RUNNING MECHANICS&COMBO MODES, to alleviate the depressive moods of the group, and to restore the inhabitants to their normal living mode.

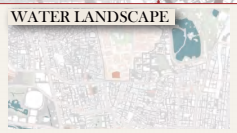
AREAS WHERE IDEAS ARE ACTIVE

MAPPING



GREEN SPACE

WATER LANDSCAPE



ROAD CONDITION

LIVING CIRCLE

SpeedNews

Urban Shelter Established

COMMONNESS

In recent years, political depression has evolved from a metaphorical symptom to a clinical condition that meets the APA criteria for depression.The structure strongly and unilaterally affects the fate of the individual, the individual can only fall into a hopeless situation.

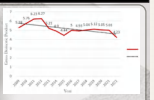
SOCIAL MALIGNANCY

THE SILENT CRY OF SOCIETY

THE BIRTH OF POLITICAL DEPRESSION

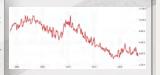
CURRENT POLITICAL EVENTS

More and more extreme political events are causing anxiety,even,if these events aren't relevant to ordinary people.



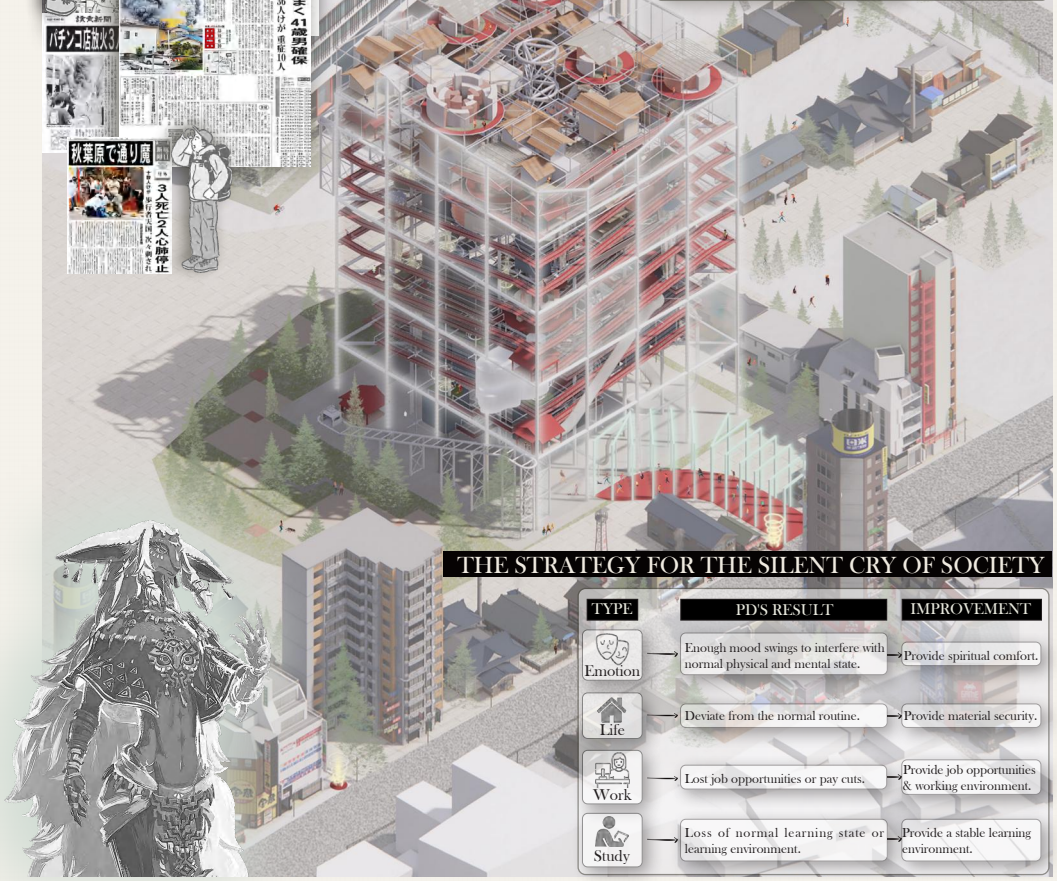
JANPANESE ECONOMY

Janpan's GDP has not been good in recent years.
The income & quality of life of ordinary people have been negatively affected.



UNEMPLOYMENT IN JAPAN

Janpan's unemployment rat,once on a downward trend,has become more volatile this year.This means that unemployment has increased in recent years.

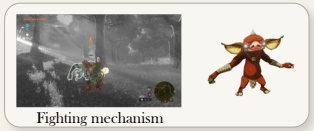


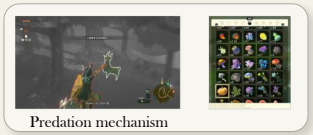
THE STRATEGY FOR THE SILENT CRY OF SOCIETY

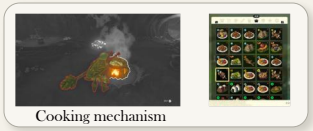
TYPE	PD'S RESULT	IMPROVEMENT
Emotion	Enough mood swings to interfere with normal physical and mental state.	Provide spiritual comfort.
Life	Deviate from the normal routine.	Provide material security.
Work	Lost job opportunities or pay cuts.	Provide job opportunities & working environment.
Study	Loss of normal learning state or learning environment.	Provide a stable learning environment.

GAME OBSERVATION

GAME MECHANICS

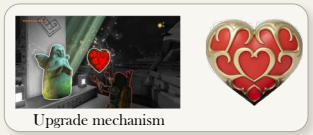

Fighting mechanism


Predation mechanism

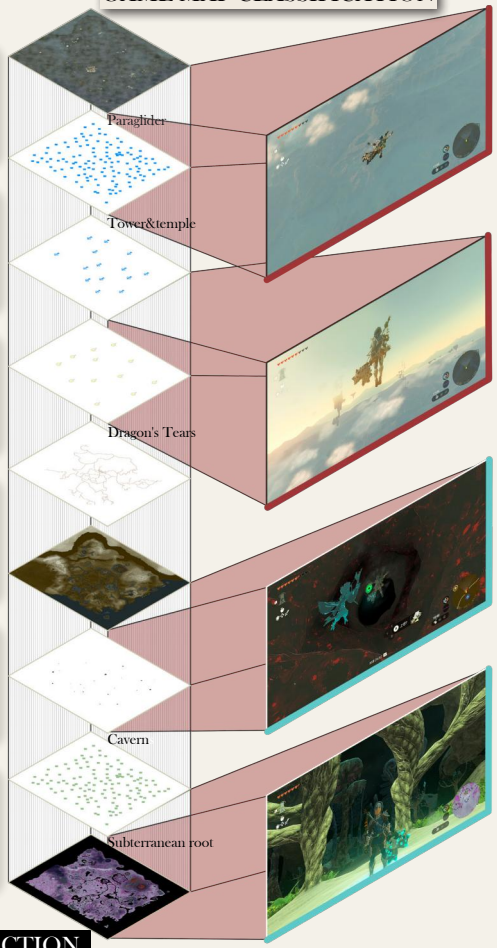

Cooking mechanism

Combined mechanism

Railcar mechanism


Upgrade mechanism

GAME MAP CLASSIFICATION



MODE OF TRANSPORT

Link jumps from an island in the air and uses a paraglider or dive function to land at the target site.

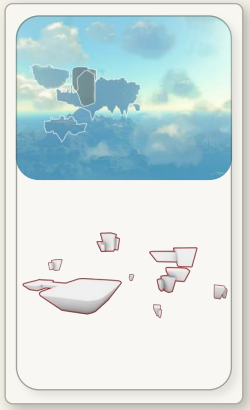
Link is connected to the island in the sky through the tower. The tower and temple also serve as a medium for Link to move quickly to target locations on the ground.

The tears of the dragon are scattered on the ground in the form of patterns, which not only explains the background of the game's story, but also promotes the development of the game's plot.

The surface is dotted with miasmatic underground caverns through which Link can enter the underground world.

Underground roots are a tool for Link to quickly move to the target location in the underground world, and it is also a tool for the underground world to connect with the outside world.

GAME PROTOTYPE EXTRACTION

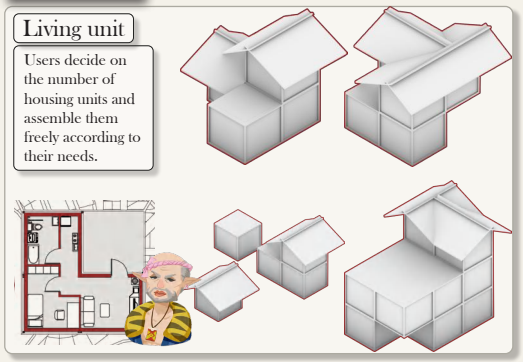


DESIGN STRATEGY

UNIT TYPE

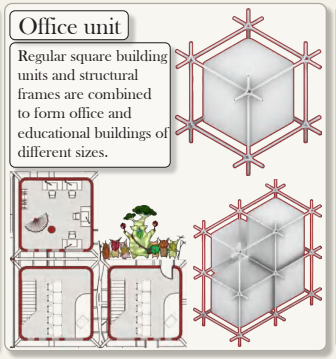
Living unit

Users decide on the number of housing units and assemble them freely according to their needs.



Office unit

Regular square building units and structural frames are combined to form office and educational buildings of different sizes.

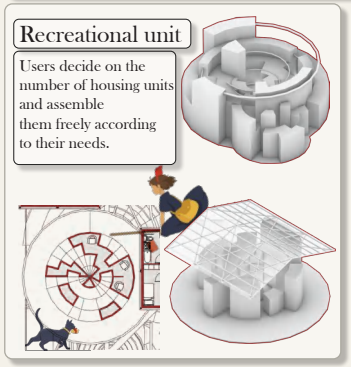


Park

Construct

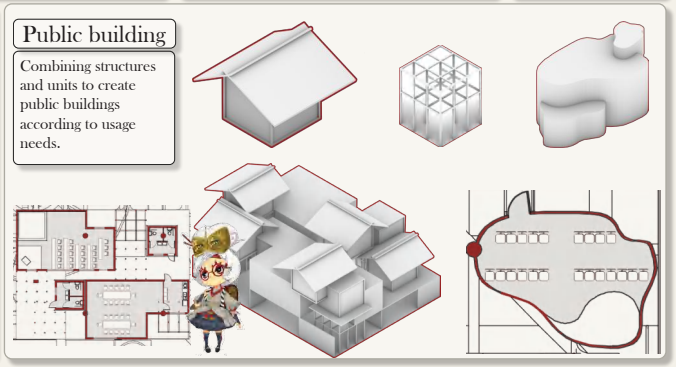
Recreational unit

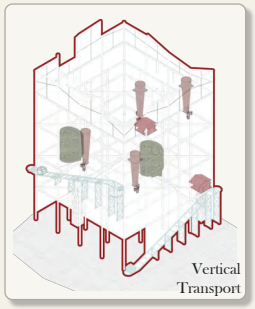
Users decide on the number of housing units and assemble them freely according to their needs.

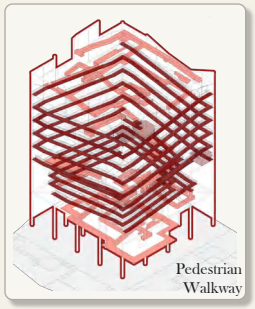


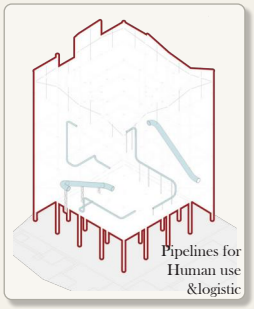
Public building

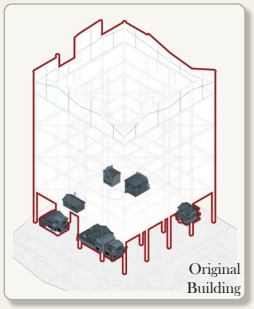
Combining structures and units to create public buildings according to usage needs.

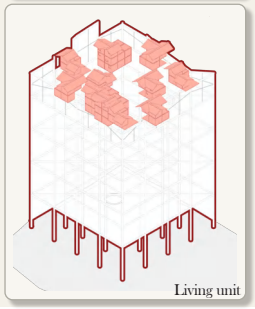


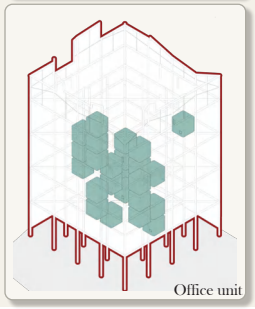

Vertical Transport

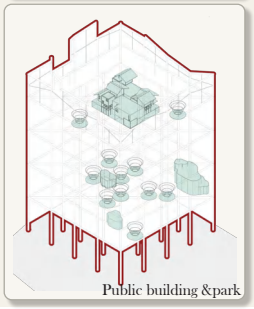

Pedestrian Walkway

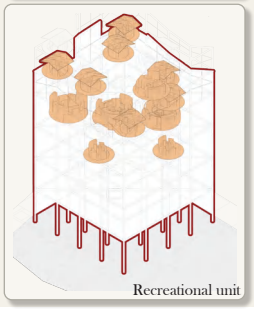

Pipelines for Human use &logistic


Original Building


Living unit

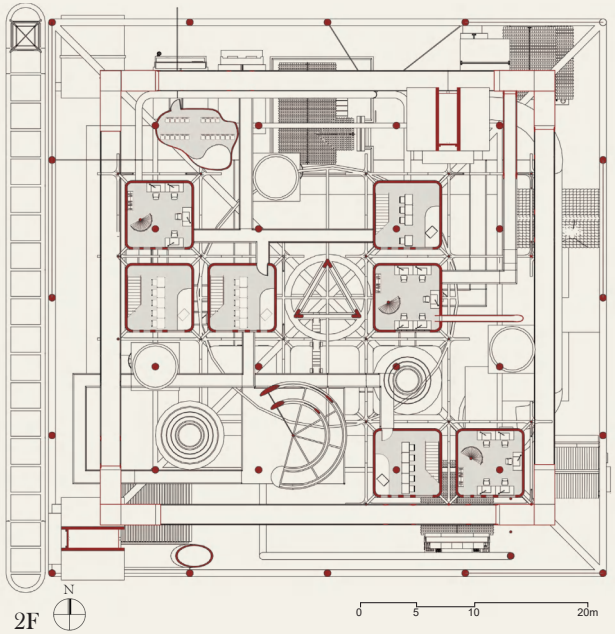

Office unit

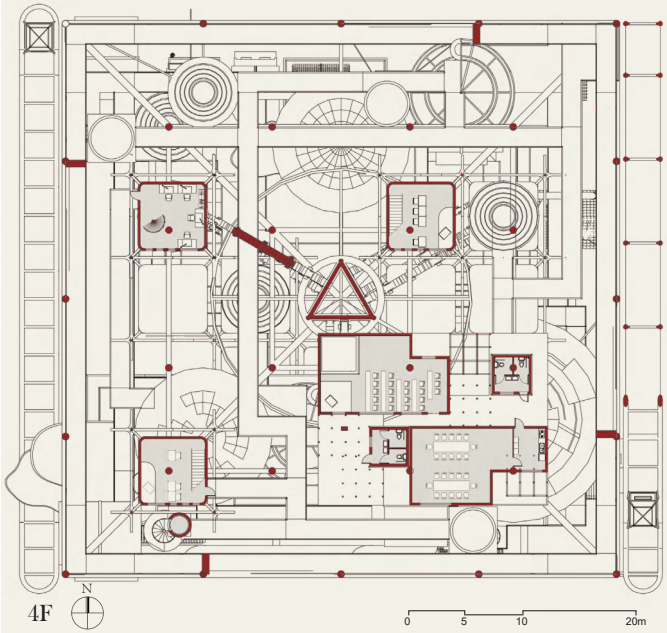

Public building &park

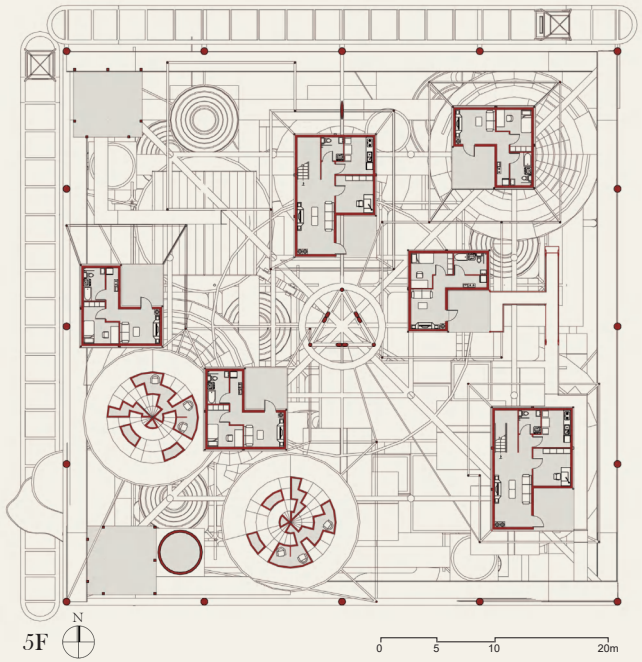

Recreational unit

MASTER PLAN

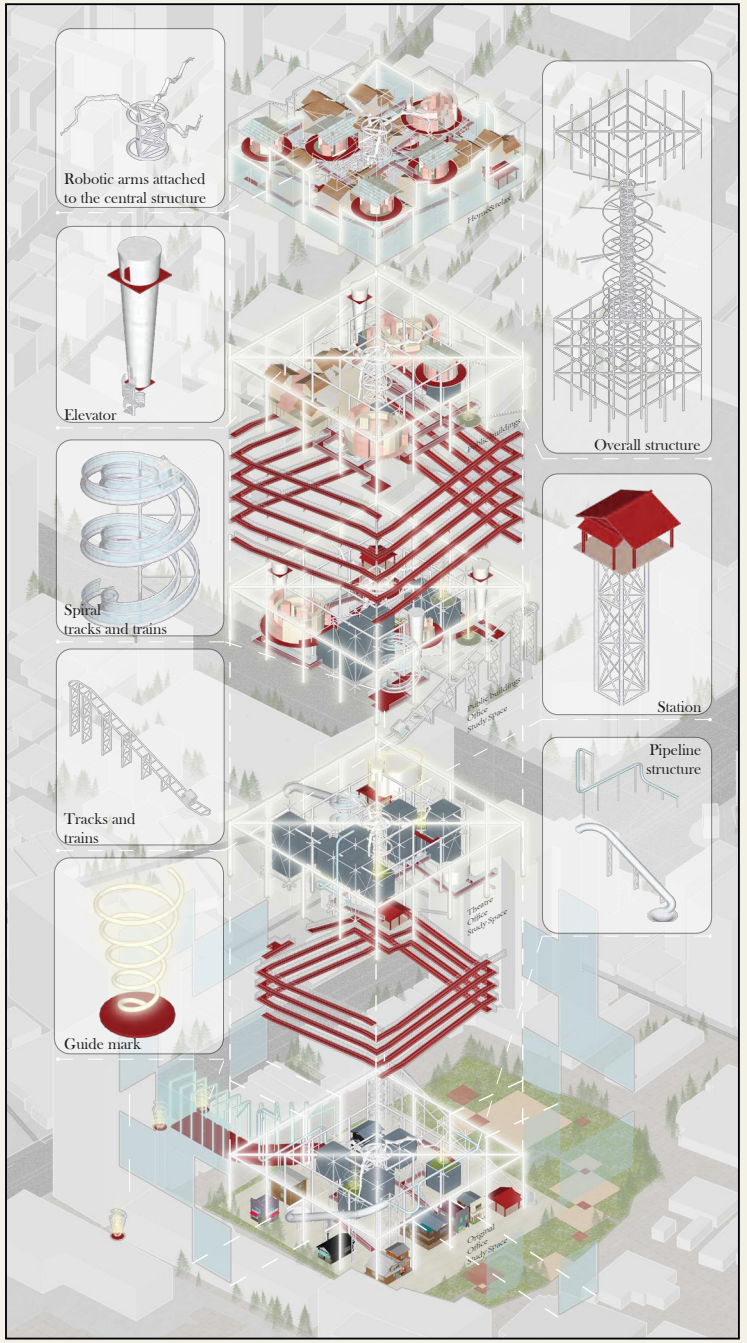
FLOOR PLAN







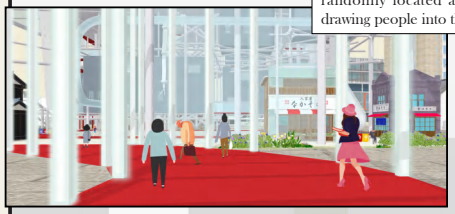
STRUCTURE&TRANSPORTATION MODE



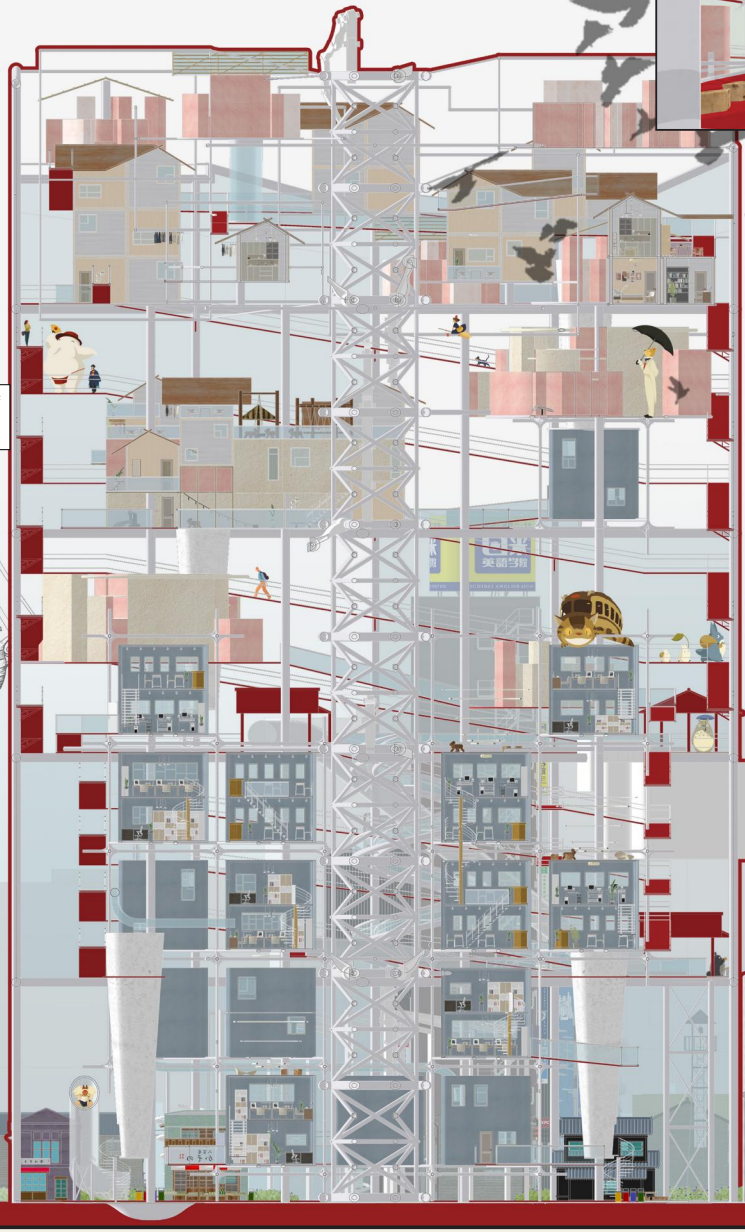
Life in urban shelter

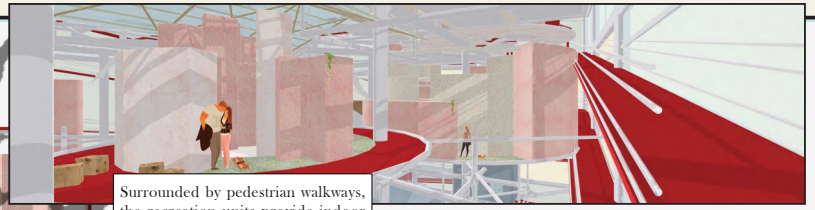
Urban shelter profile

Glowing, mysterious constructs are randomly located around the sanctuary, drawing people into the city's shelter.

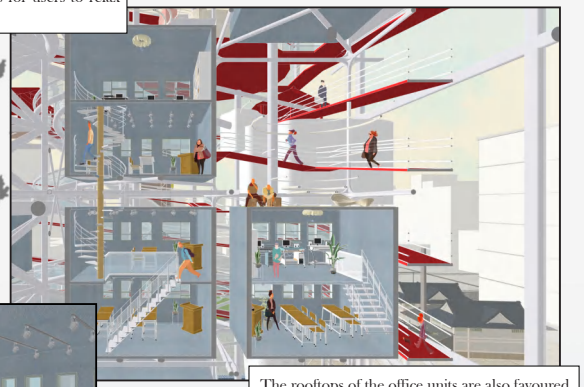


Entering the first floor of the urban refuge, the retained buildings of the original site are visible along with the refuge buildings that invaded the original site, with the first floor dominated by office buildings.





Surrounded by pedestrian walkways, the recreation units provide indoor and outdoor spaces for users to relax and unwind.

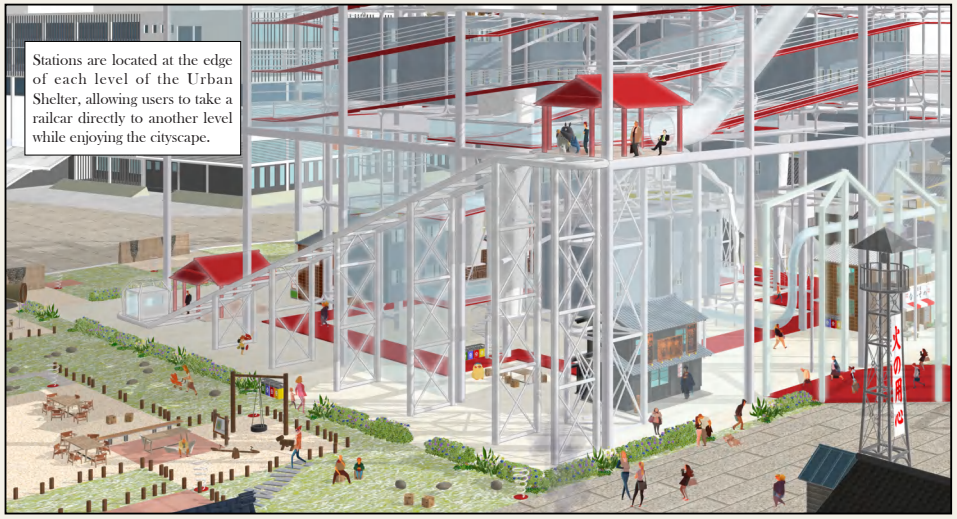


The rooftops of the office units are also favoured by the users for outdoor study and relaxation.

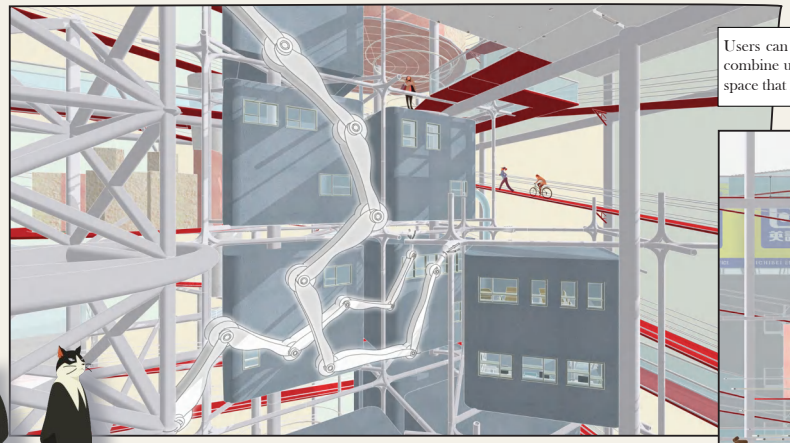
The combined office units reflect the school's attributes, and users who are teachers and students are provided with teaching space.

Users can teach, learn and work in the office units.

Users with children can send them to kindergarten in city shelter. In the kindergarten, children are offered a wide range of sports, recreational activities and more.

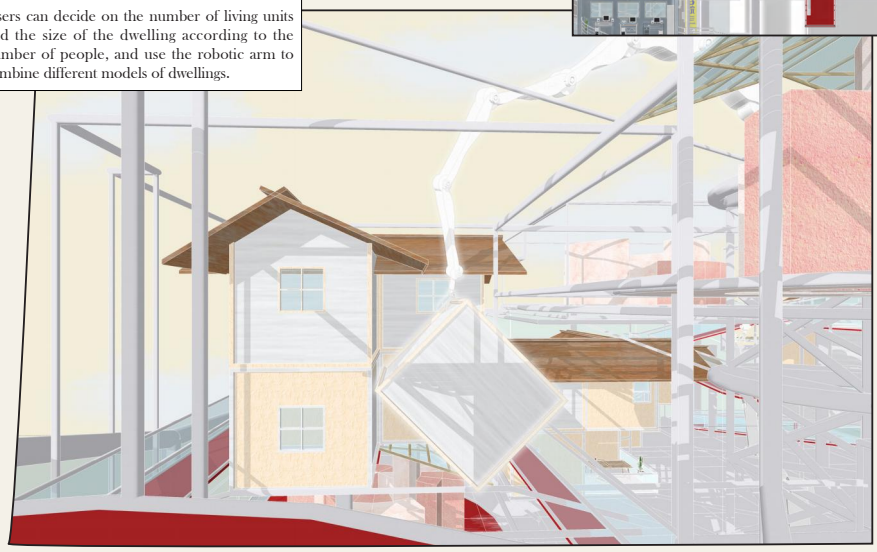


Stations are located at the edge of each level of the Urban Shelter, allowing users to take a railcar directly to another level while enjoying the cityscape.

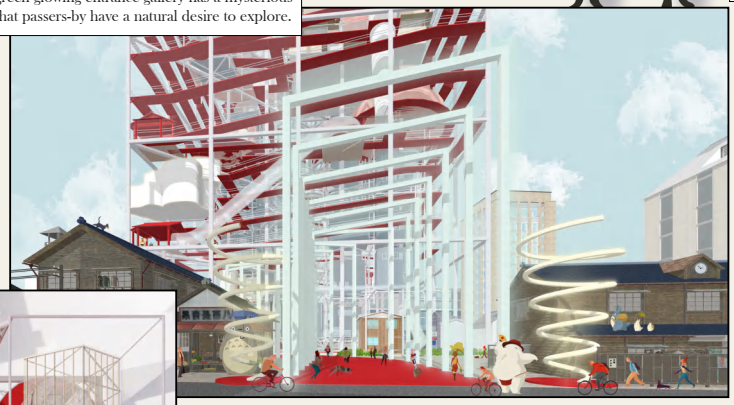


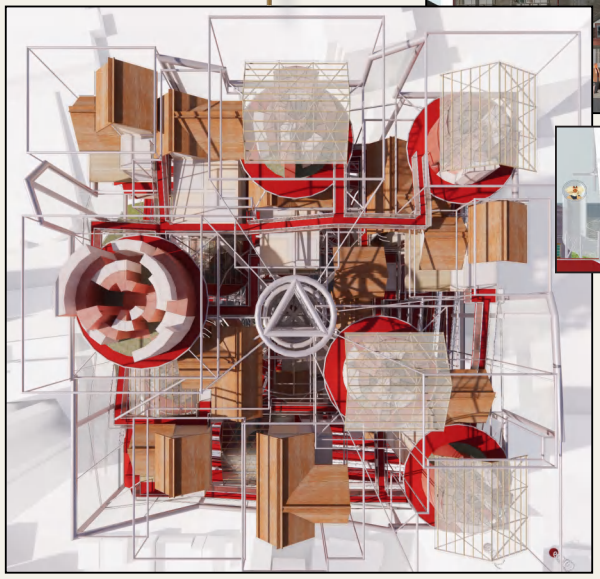
Users can operate the robotic arm to freely combine units of the same nature to create a space that meets their individual needs.

Users can decide on the number of living units and the size of the dwelling according to the number of people, and use the robotic arm to combine different models of dwellings.

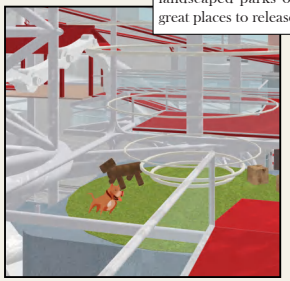


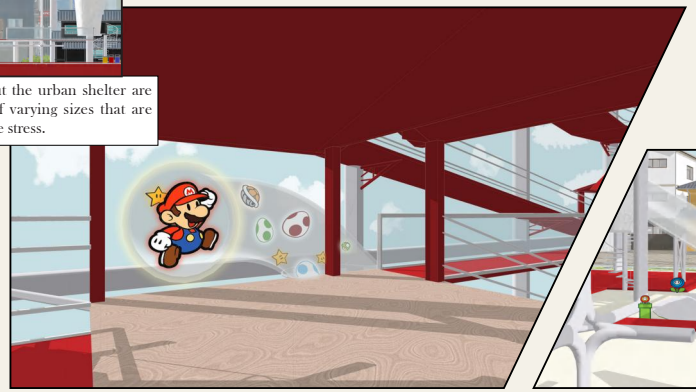
The blue-green glowing entrance gallery has a mysterious character that passers-by have a natural desire to explore.

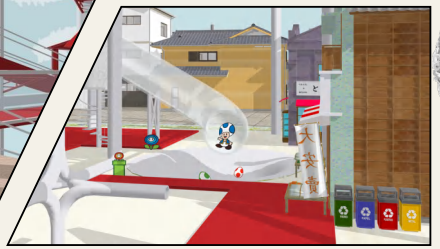




Scattered throughout the urban shelter are landscaped parks of varying sizes that are great places to release stress.







The stations in Urban Shelter are also equipped with highly interesting pipes for users to quickly reach their destinations. Miniature-sized tubes have also been installed between office units for the transfer of items.

02 INVASION BY SLIME MOULD

RENEWAL OF ABANDONED METRO SPACE

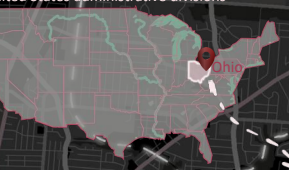
Location:Cincinnati, Ohio, United States of America
Area:683,099 m² Duration:2024.10-11
Type:Urban design Academic individual work

Cincinnati had plans to convert a river into an underground station, but due to the negative effects of the World Wars, the Cincinnati government did not have the financial resources to continue with the planned underground, leaving unfinished underground spaces that remain to this day. These **NEGATIVE UNDERGROUND SPACES** are used by daredevils and graffiti artists, and provide a breeding ground for crime in the city, and are in dire need of renovation.

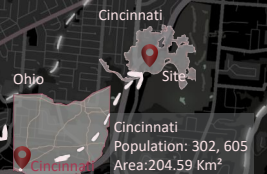
This project uses the **SLIME MOULD ALGORITHM** in grasshopper to generate **ARCHITECTURAL ENCROACHMENTS** into abandoned urban spaces for the purpose of **SPATIAL RECOVERY** and **URBAN REGENERATION.**

CINCINNATI'S HISTORIC POPULATION

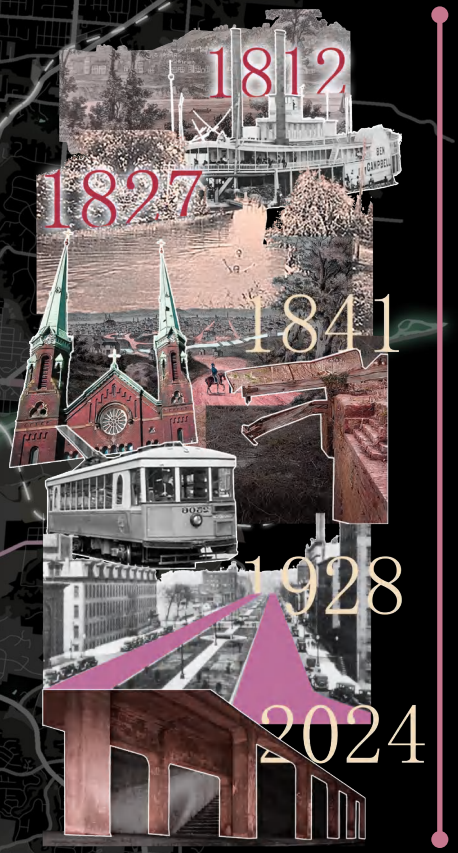
United States administrative divisions



CITY EVOLUTION



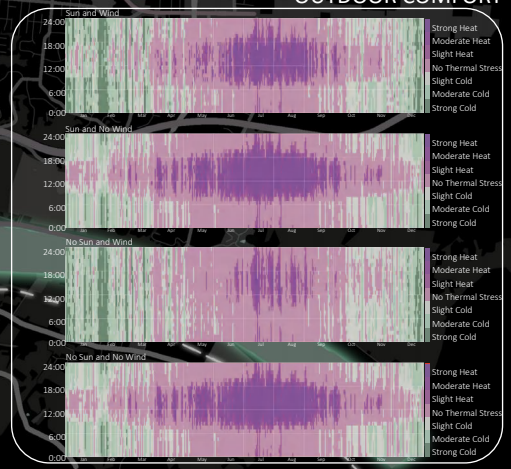
HISTORICAL DEVELOPMENT



TRAFFIC ANALYSIS

ETHNIC DISTRIBUTION

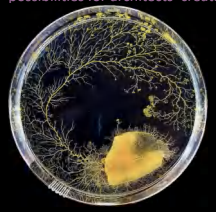
OUTDOOR COMFORT



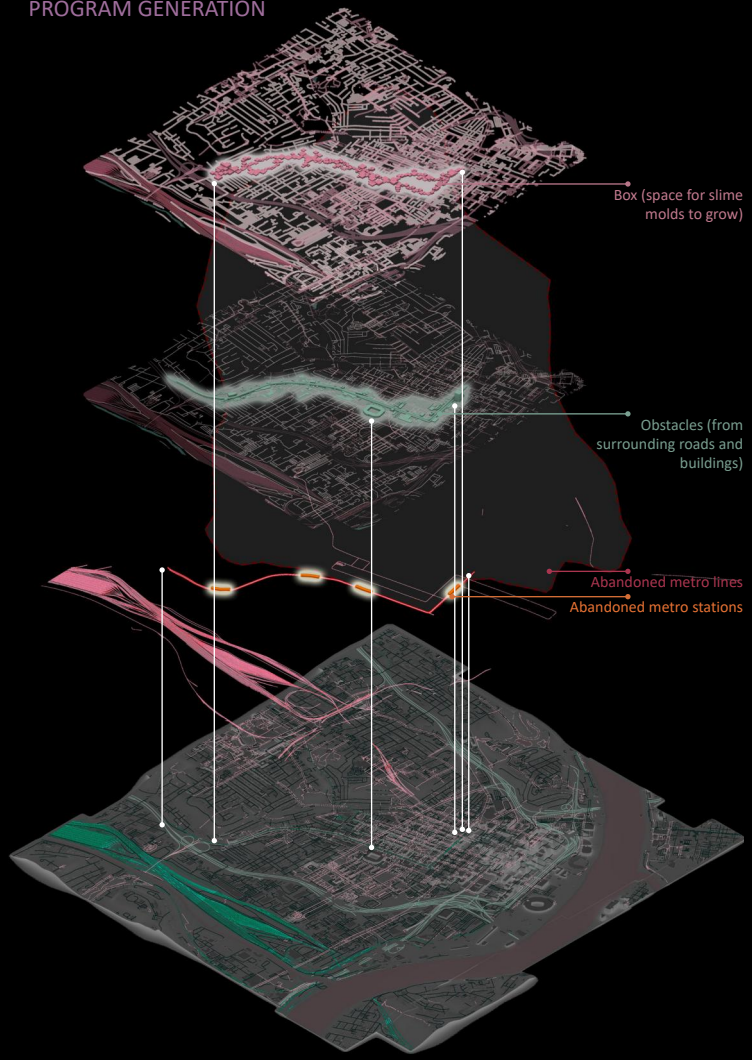
SLIME MOULD ALGORITHM

As a eukaryotic organism, slime moulds appear similar to animals, protozoa, fungi or plants throughout their life cycle. Although they do not have a nervous system, they show even more complex behaviours during foraging, and the network that builds the foraging will spend the shortest time possible to draw food in the most efficient way.

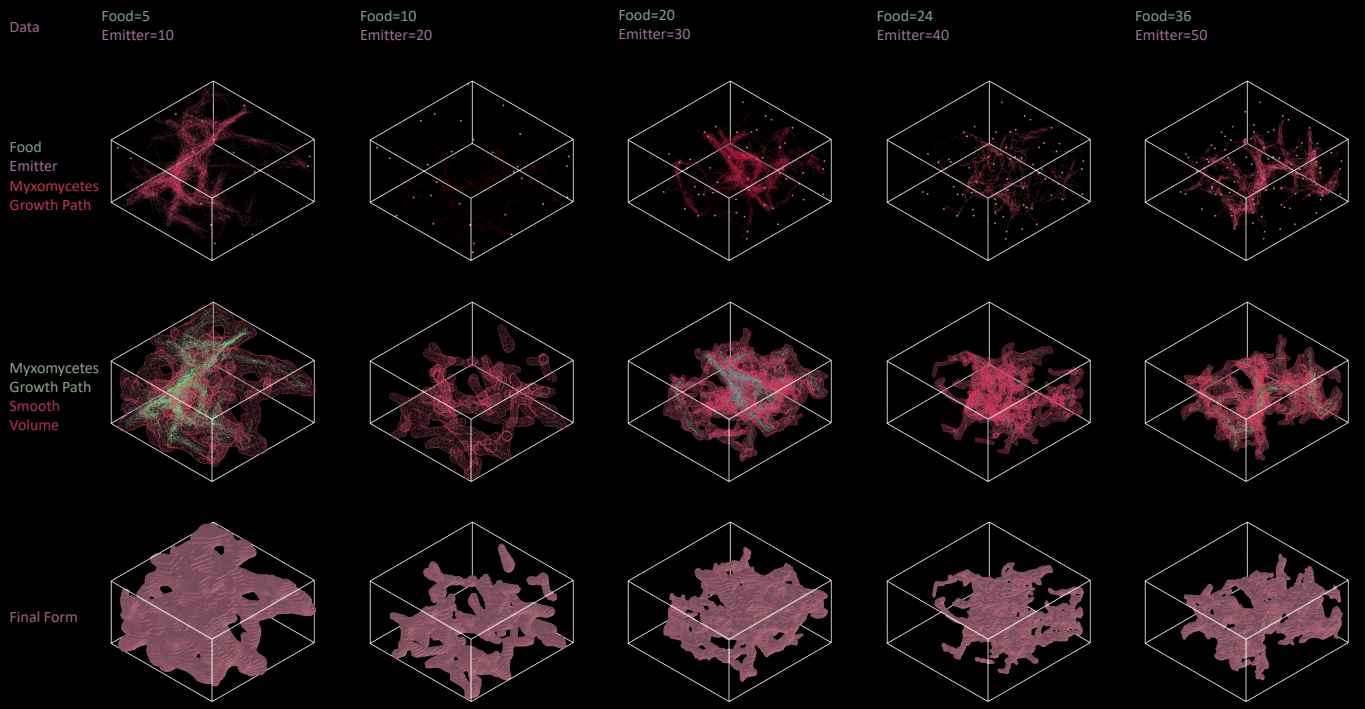
Algorithms based on the simulation of slime mold behaviour make the parametric model optimized by this system more diversified in terms of internal spatial organisation, bringing rational references as well as more possibilities for architects' creations.



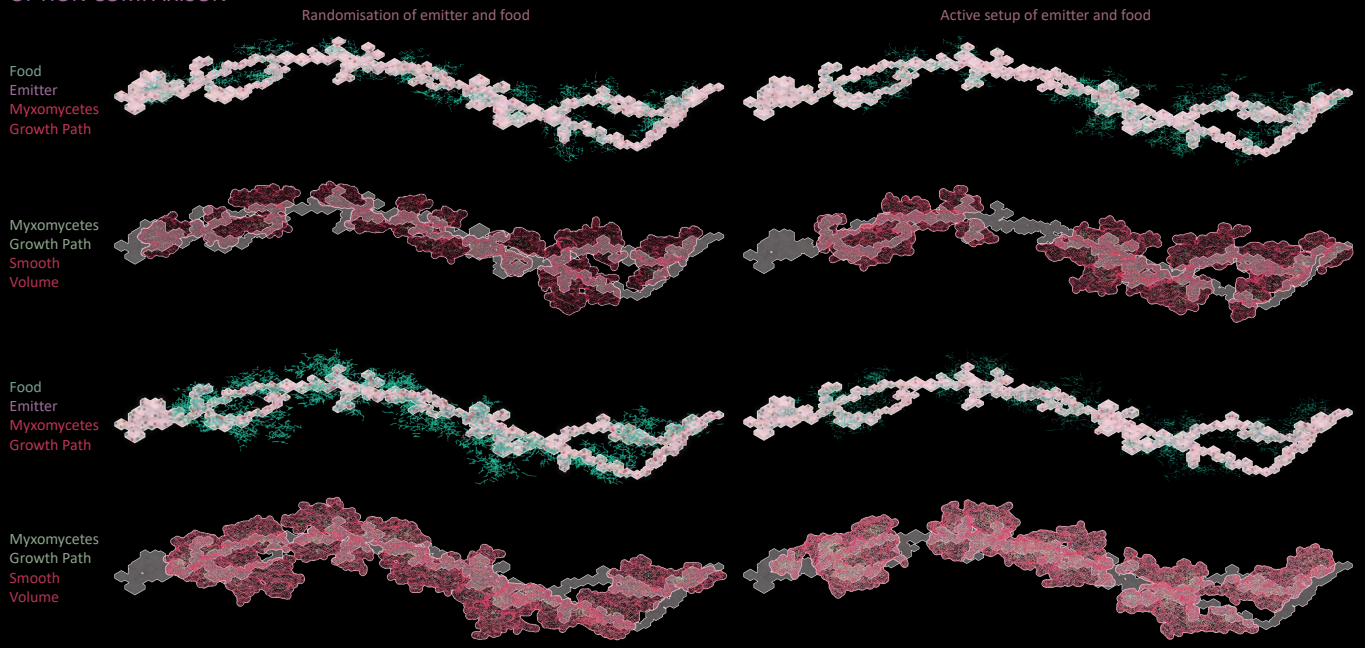
PROGRAM GENERATION

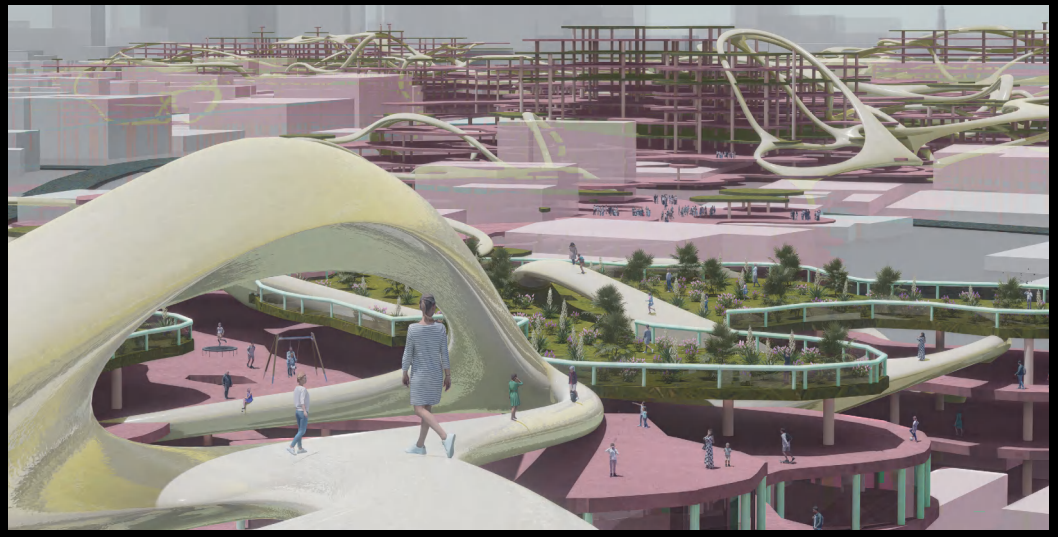


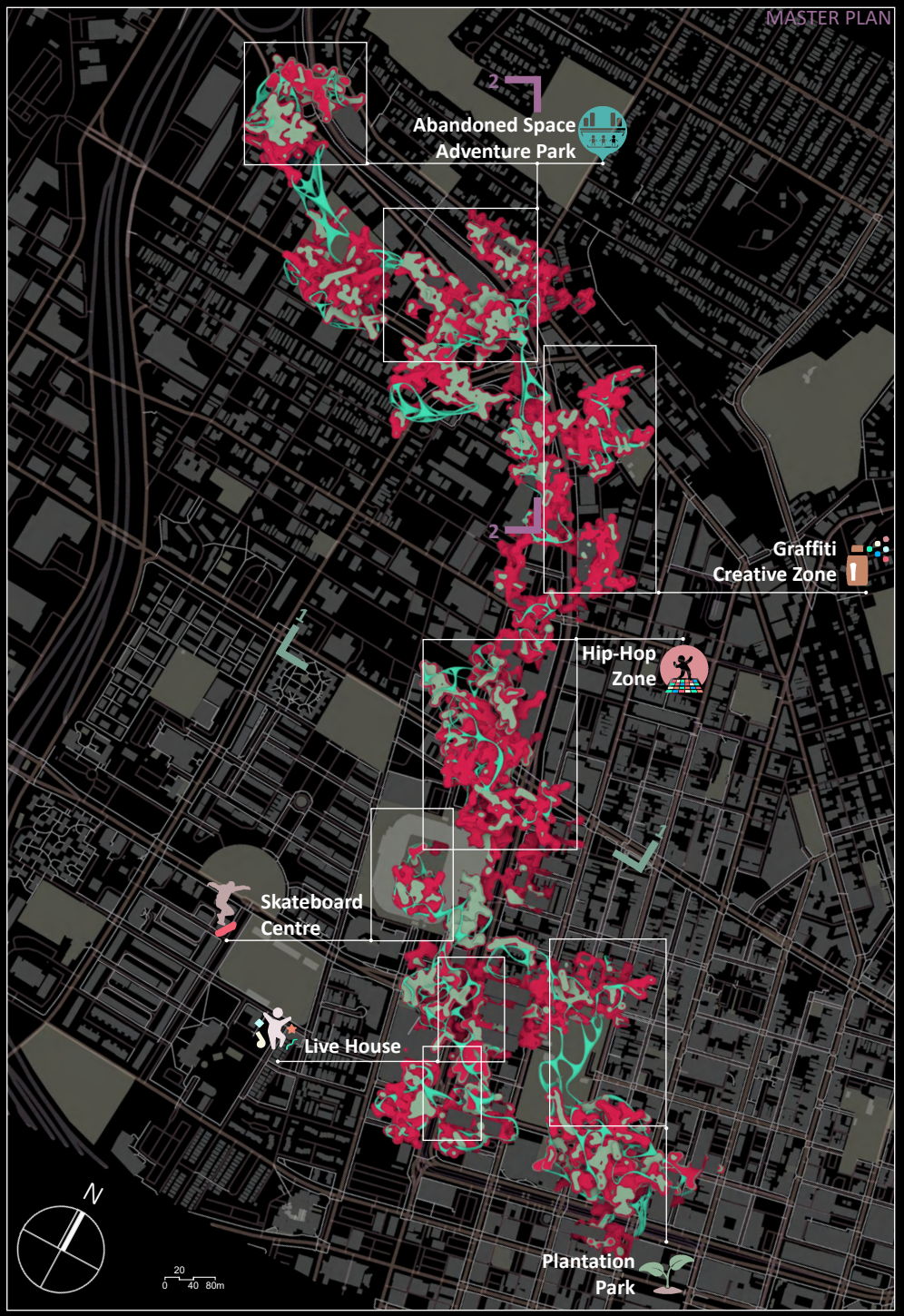
SCHEMATIC DIAGRAM OF THE SLIME MOULD ALGORITHM



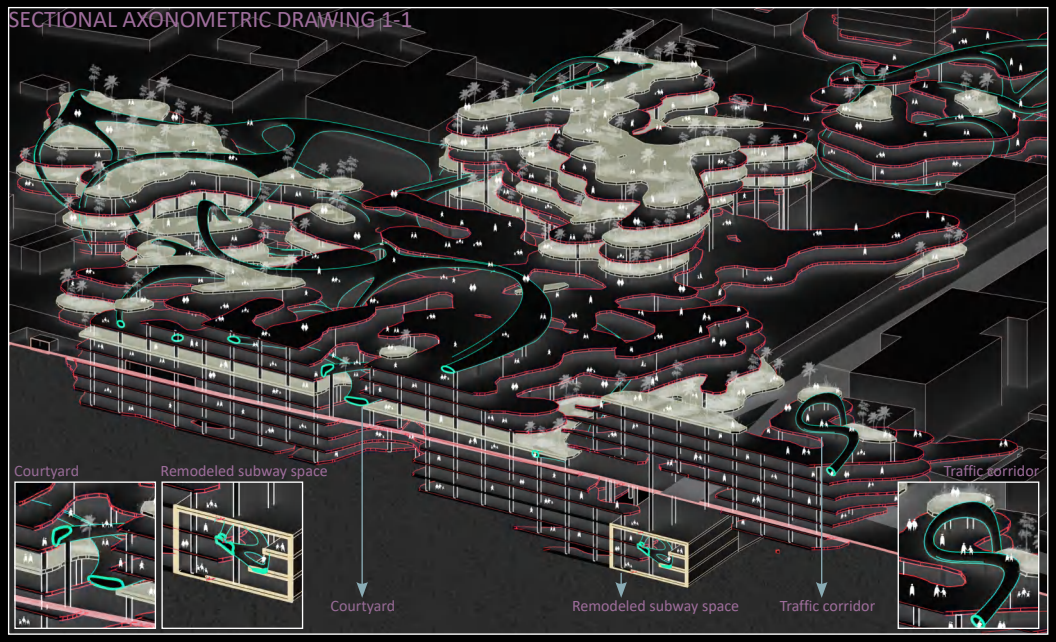
OPTION COMPARISON



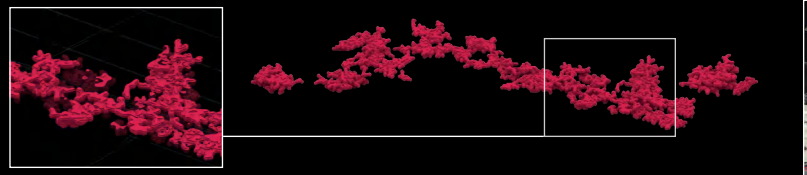



MASTER PLAN
2
Abandoned Space
Adventure Park
2
2
1
Graffiti
Creative Zone
Hip-Hop
Zone
1
Skateboard
Centre
Live House
N
20
0 40 80m
Plantation
Park

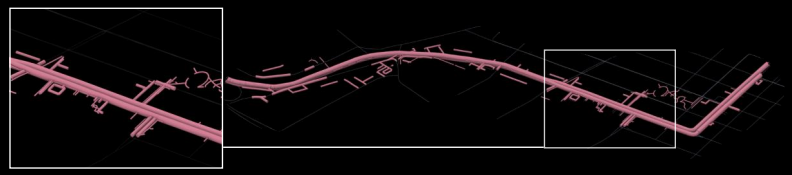
SECTIONAL AXONOMETRIC DRAWING 1-1



FUNCTIONAL LAYOUT Invasive building

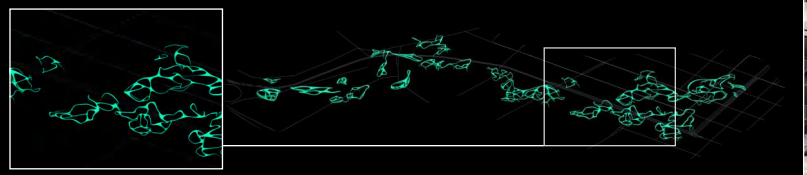


Uninvaded space

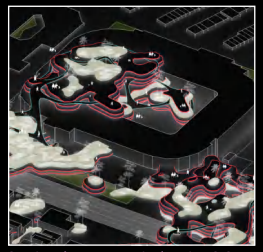


Invaded buildings

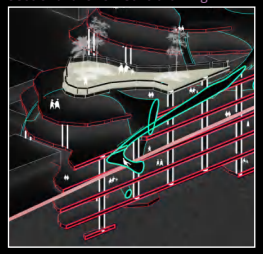
Traffic space



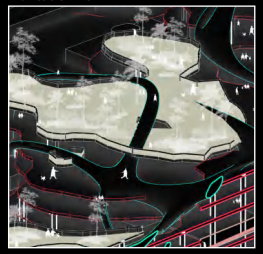
Slime Mould invasion of the gym

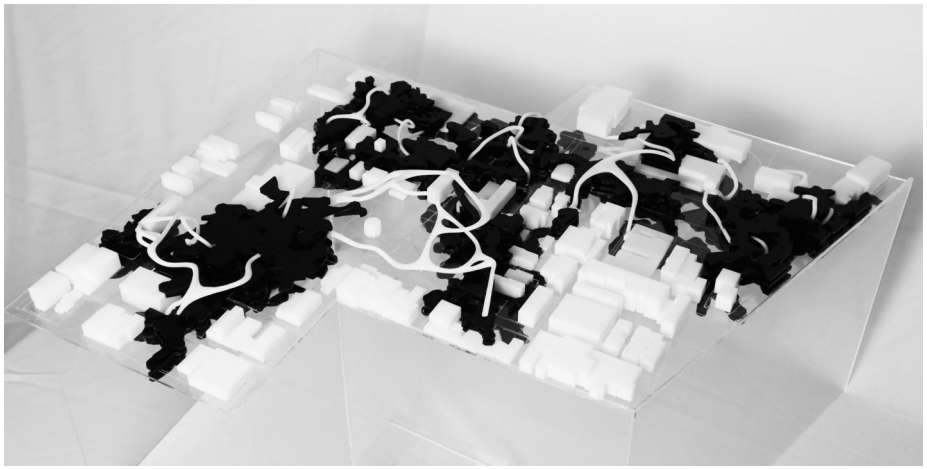


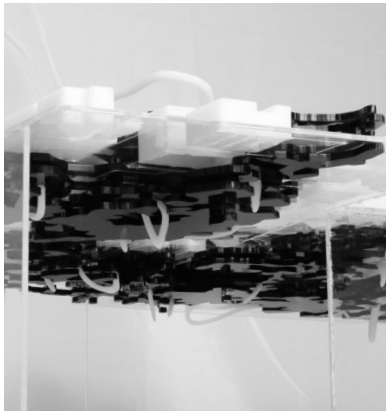
Sectional axonometric drawing 2-2

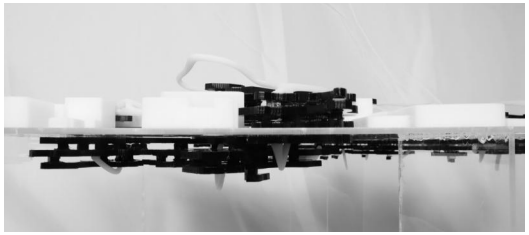


Plantation Park









MANUAL MODEL

Size:45.2*51.4*23cm
Material:Acrylic sheet;white resin

PRODUCTION PROCESS

White acrylic is used to represent the original buildings, black acrylic is used to represent the buildingswWWWW generated according to the slime mold algorithm, and transparent acrylic represents the ground level, which makes it easy to see the underground parts.

03 NAIROBI CYCLE

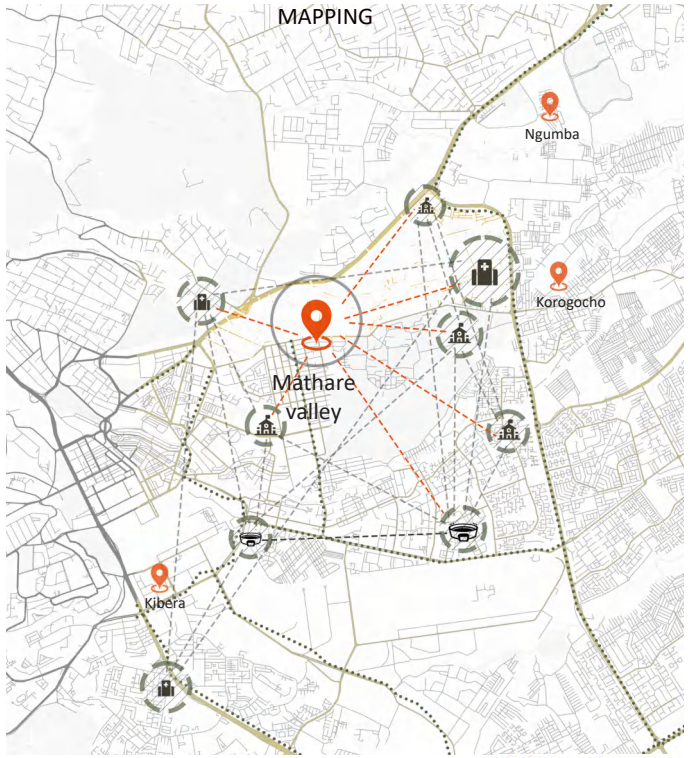
SLUM REHABILITATION

Location:Nairobi, Kenya Area:332,000m²
Duration:2024.7-9 Type:Urban design
Designer:yumeng cheng,hanyun hu.baiyi wang, xiaoyu wang.
My Role:aboriginal population analysis,region 2 design ,Overall site operating model design.
Design component:region 2

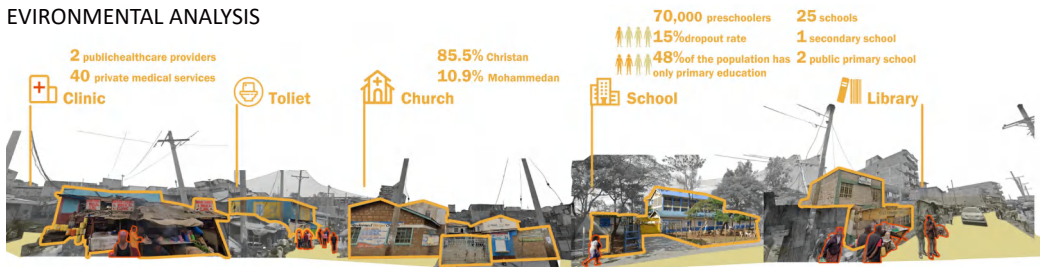
The Mathare Valley is one of the oldest and used to be one of **THE WORST SLUM AREAS** in Nairobi.Most of the residents here don't work or go to school, and the youth are addicted to drug-like vices. People live in **6 FT.*8 FT.SHANTIES** made of old tin and mud.There are not furniture,no utility system.There are public pay toilets shared by up to 100 people."Flying toilets"were made available to poorer groups, but this meant that they would dispose of their faeces in the street or in the cars of passers-byat night, or even in the Nairobi River,which is the residents'water supply.

NAIROBI CYCLE transforms slums into new neighbourhoods by designing **A COMMUNITY SELF-FUNCTIONING MODEL** that involves residents in community building and plantation industries. The local wine industry will **BE REGULATED** to meet the needs of the region's business and health.

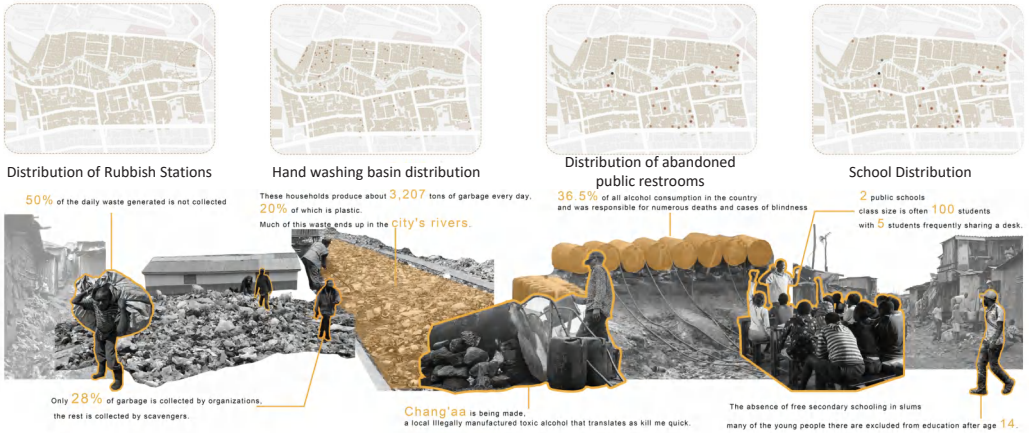
MAPPING



EVIRONMENTAL ANALYSIS



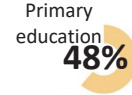
INFRASTRUCTURE SITUATIONAL ANALYSIS



SITE PROBLEM ANALYSIS

2.5million Slum dwellers in Nairobi

Primary education **48%**

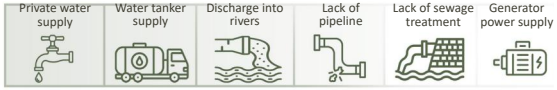


68,942/km² Highest population density

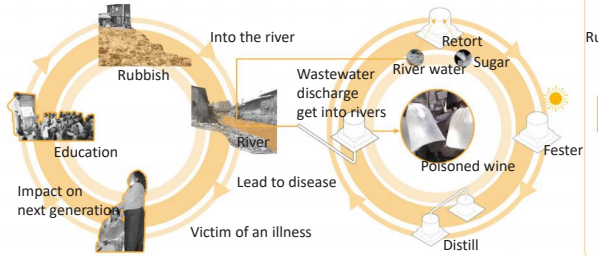
PROBLEMS IN MATHARE

PROBLEMS IN CHILDREN PROTECTION

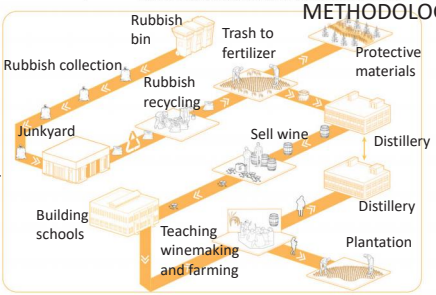
PROBLEMS IN INFRASTRUCTURE



PRODUCTION PROCESS OF TOXIC SUBSTANCES IN MATHARE

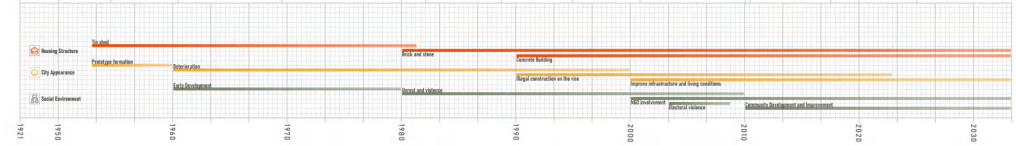


METHODOLOG

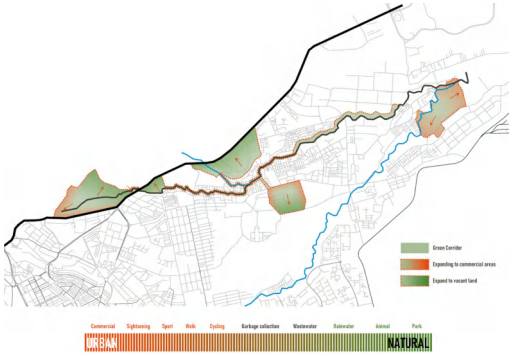


HISTORICAL ANALYSIS

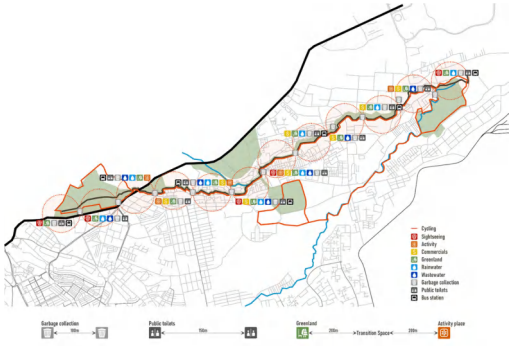
HISTORICAL EVOLUTION



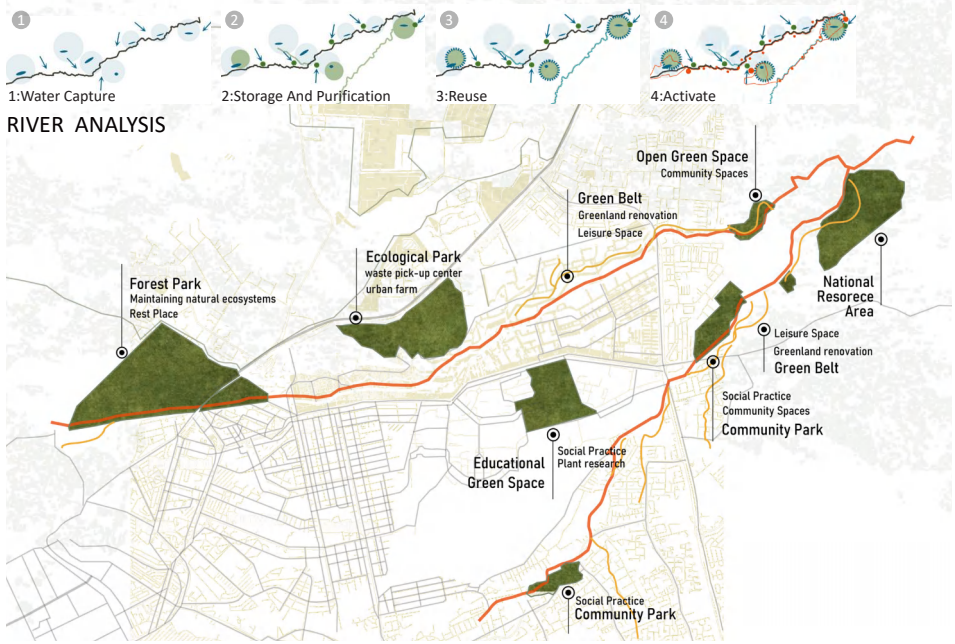
MATHARE RIVER LANDSCAPE PLAN



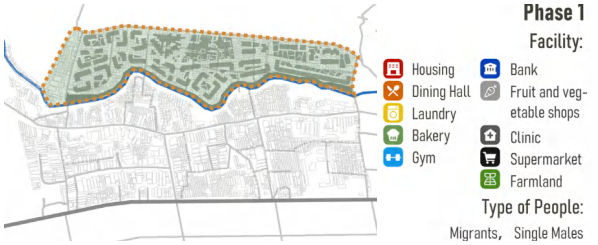
MATHARE RIVER FACILITY PLAN



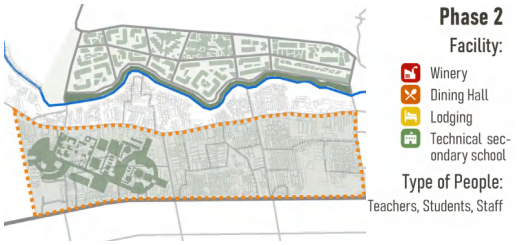
OVERALL VISION FOR THE MATHARE RIVER



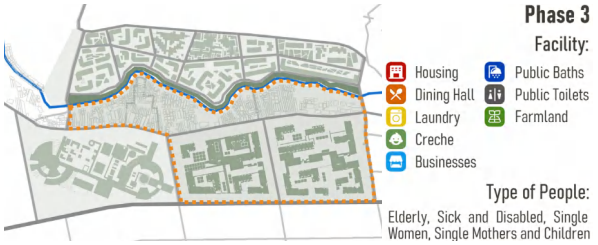
PHASED CONSTRUCTION PLAN



Phase 1 intends to demolish self-built squatters in the circled area and recycleconstruction waste, followed by the construction of new short-term accommodationareas.



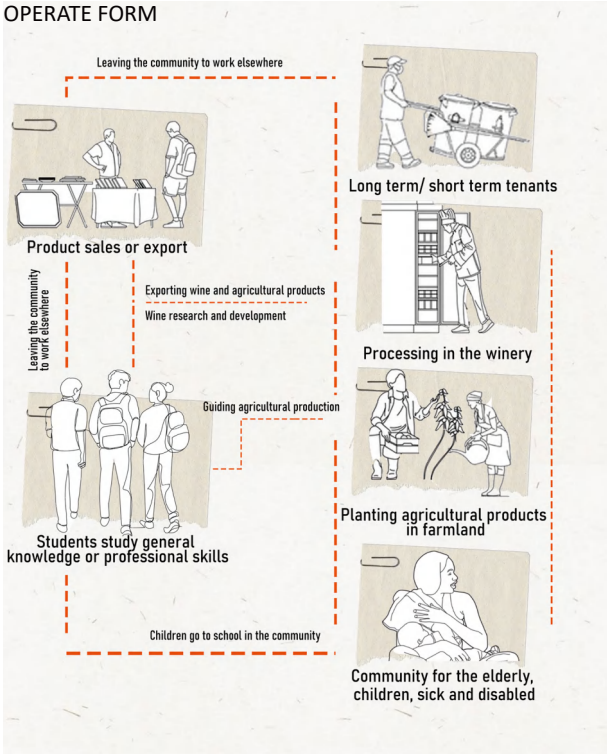
Phase 2 intends to demolish the self-built squatters in the circled area andrecycle the construction waste, which together with the waste from the firstphase will be used for the construction of a winery and a new short-termresidential area after the secondary school.



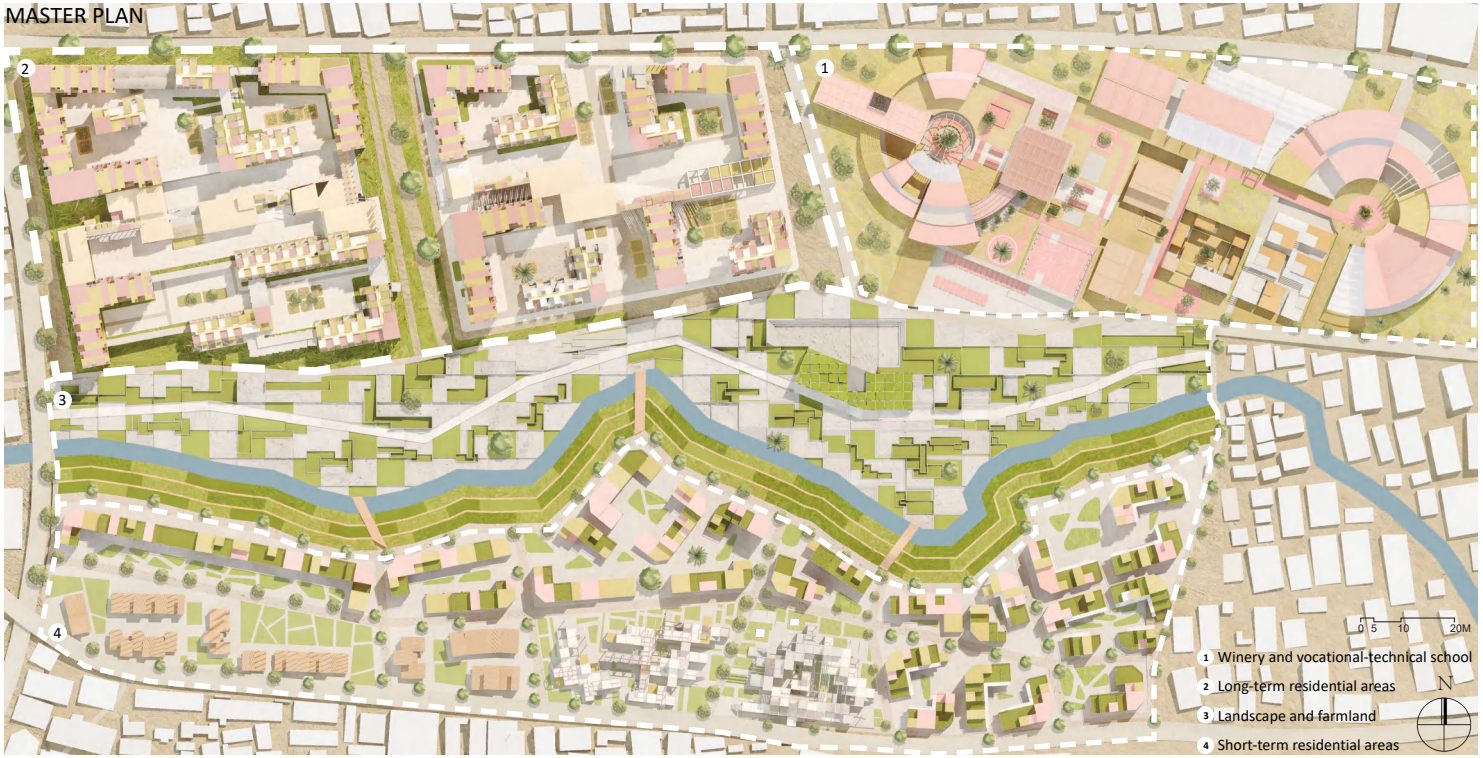
Phase 3 intends to demolish the self-constructed squatters in the circled area and builda permanent residential area using new materials.

Phase 4 intends to demolish the self-built squatters in the circled portion andbuild the landscaped and farmland portions.

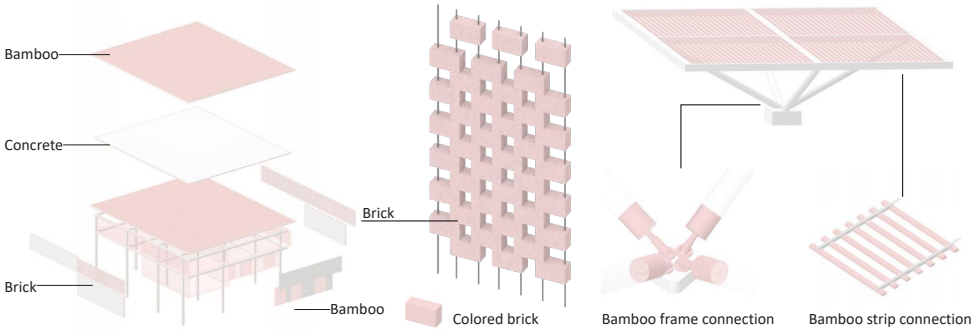
OPERATE FORM



MASTER PLAN

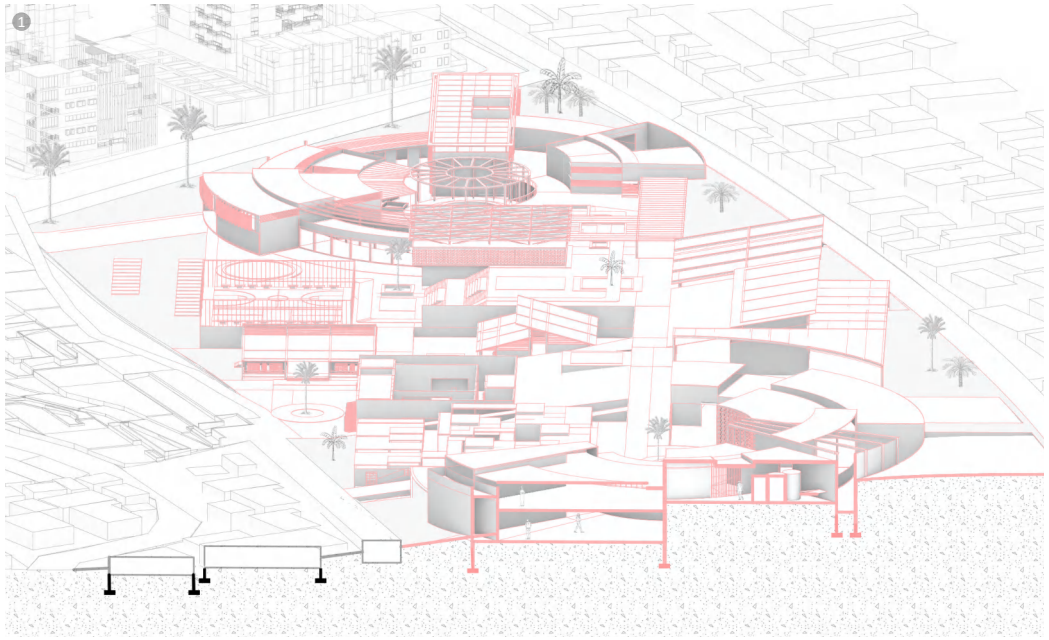


MATERIAL ANALYSIS



SPACE DISPLAY

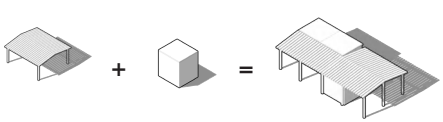
SECTION OF THE WINERY



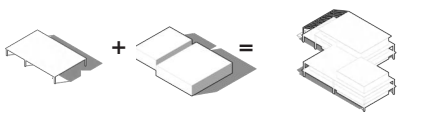
FLOOR PLAN

SHARED SPACE

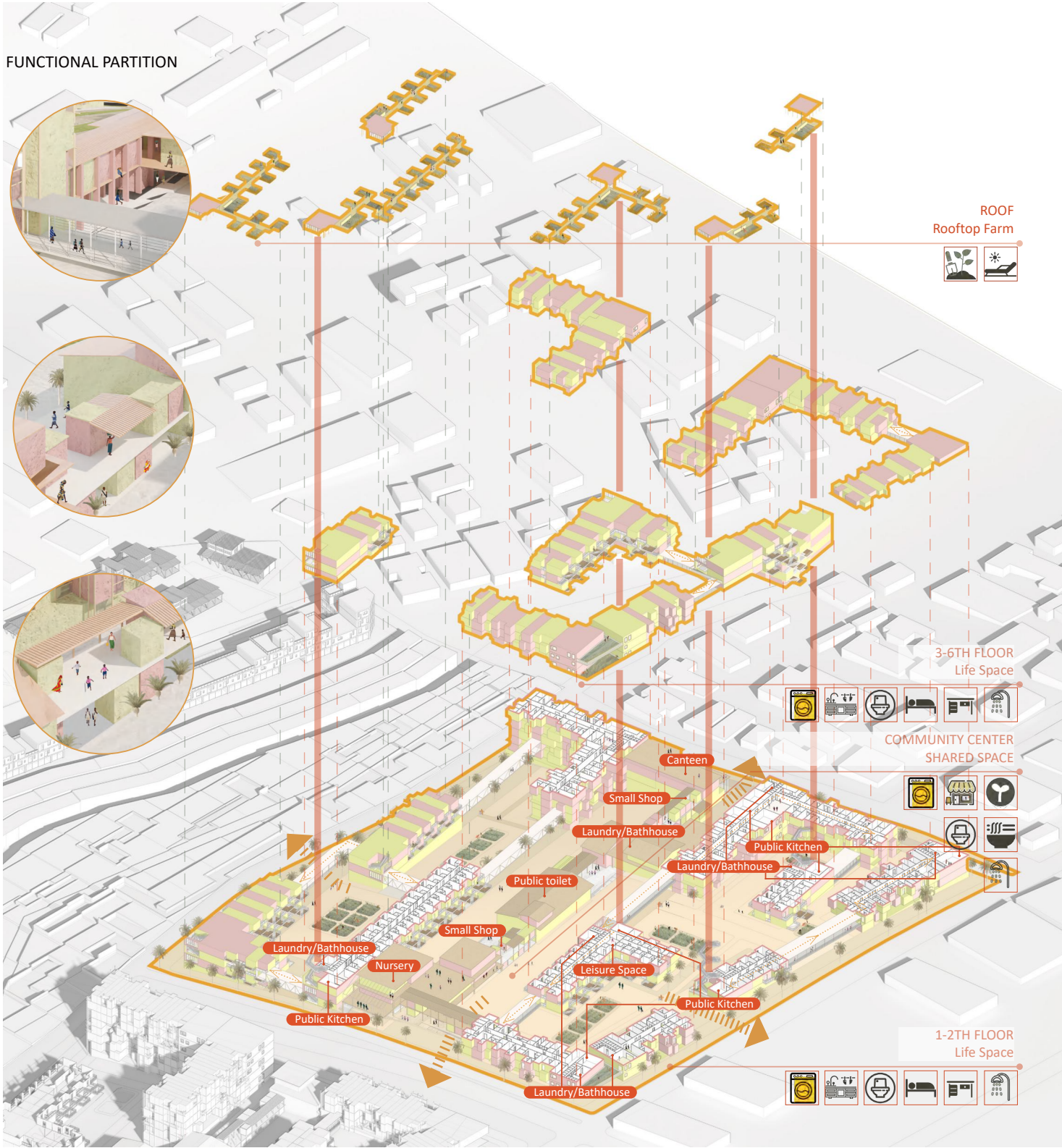
Shop&Public Bathroom

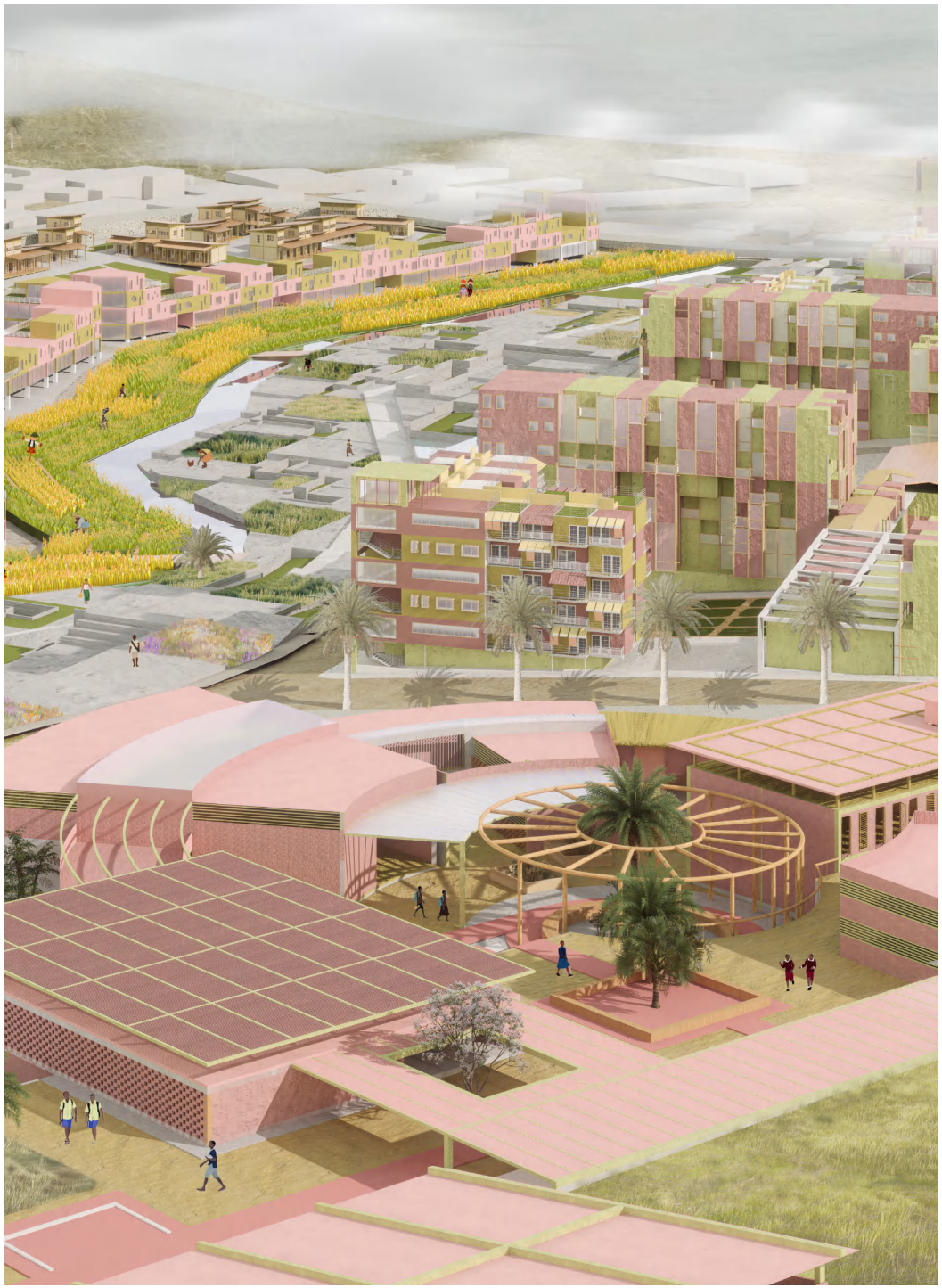


Kindergarten



FUNCTIONAL PARTITION





3-STOREY FLOOR PLAN

1:Flight of stairs

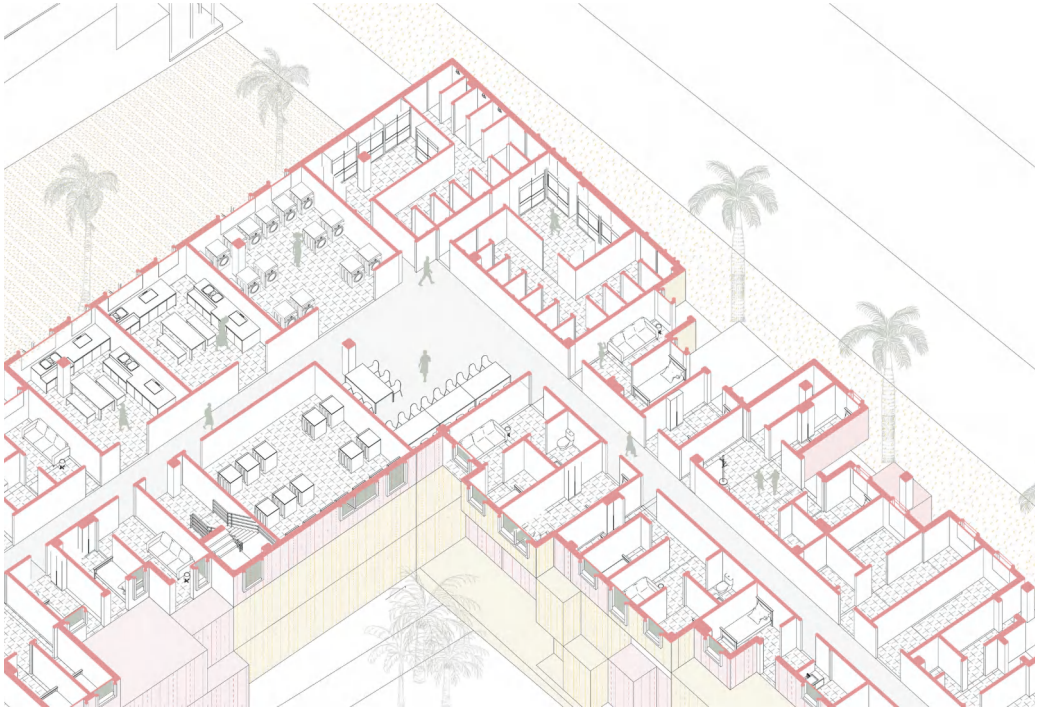
2:Public bathroom

3:Room

4:Public laundry

5:Public kitchen

6:Public activity area



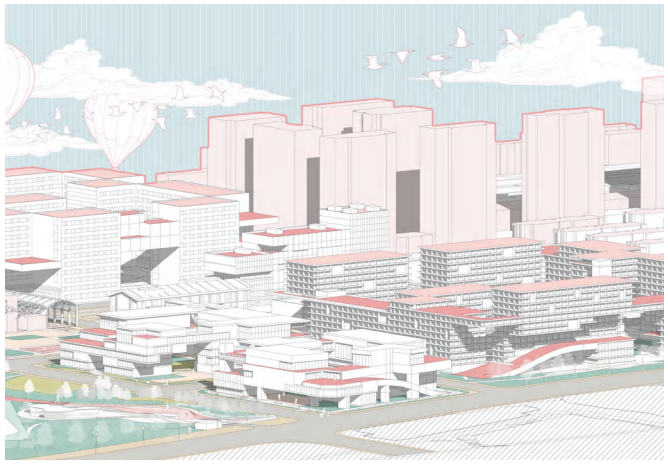
04

Urban Habitat

REHABILITATION OF INDUSTRIAL HERITAGE

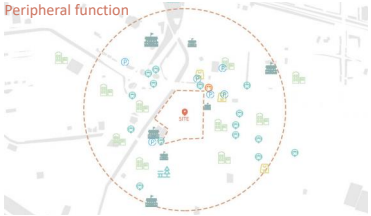
Location:Tianjin First Machine Tool Factory, Tianjin,China
Area:329,690m² Duration:2024.3-5
Type:Urban regeneration design
Academic individual work

Focusing on the two major issues of **'URBAN ANIMAL SURVIVAL CONFLICTS'** & **'URBAN OLD AGE'**, it will make a soft transition from ecological zones to habitats. Relying on the rich cultural heritage of industrial parks, selectively retain, **REBUILD** or **DEMOLISH** the original factory buildings and implant functions such as industrial heritage museums and cultural and creative centres.



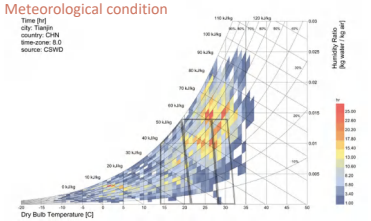
ENVIRONMENTAL CONDITION

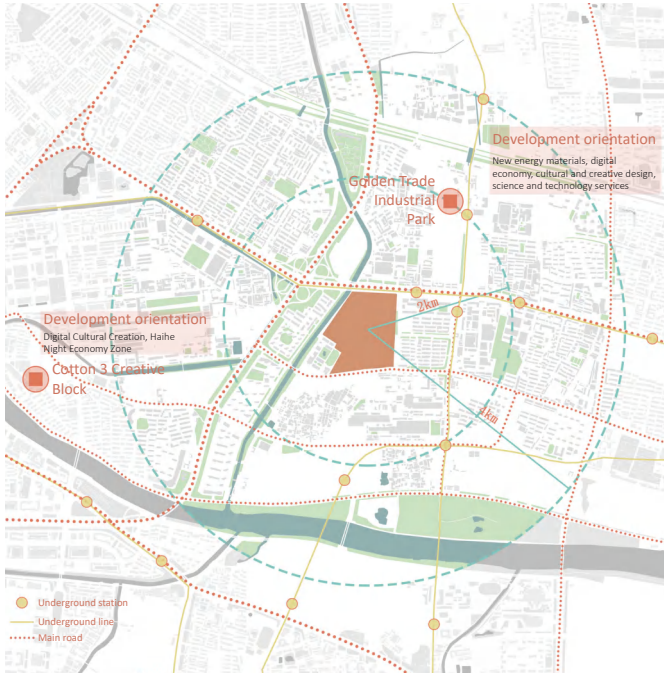
Peripheral function



Eco-space

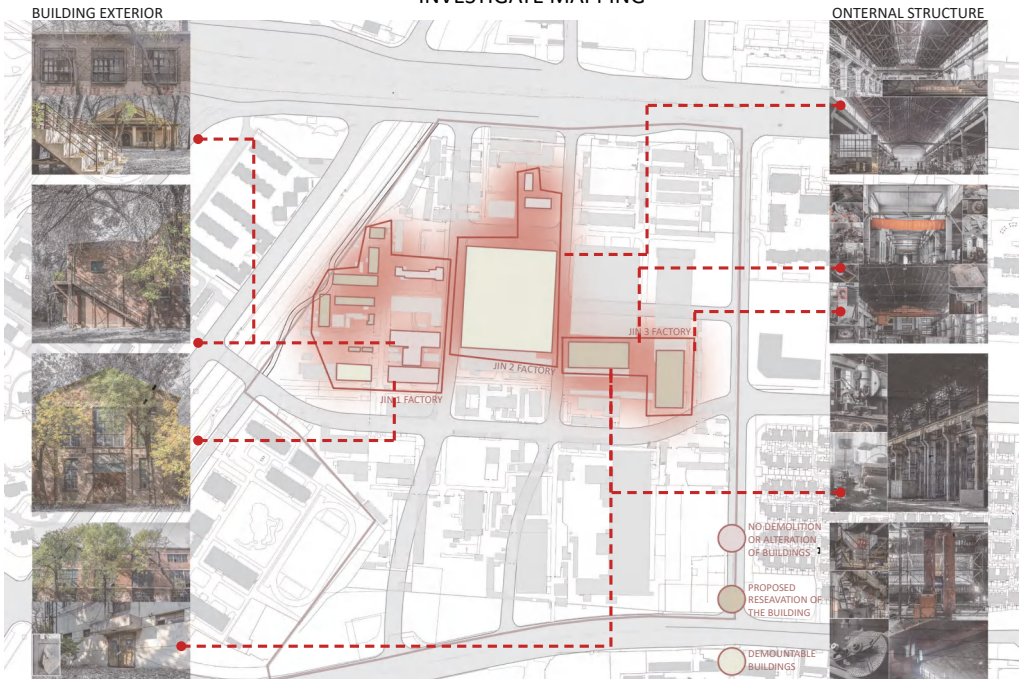
Meteorological condition

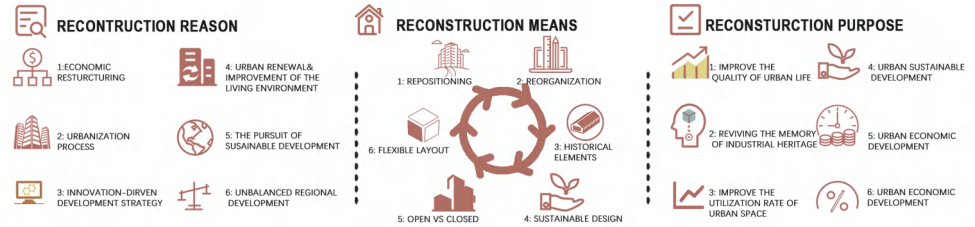




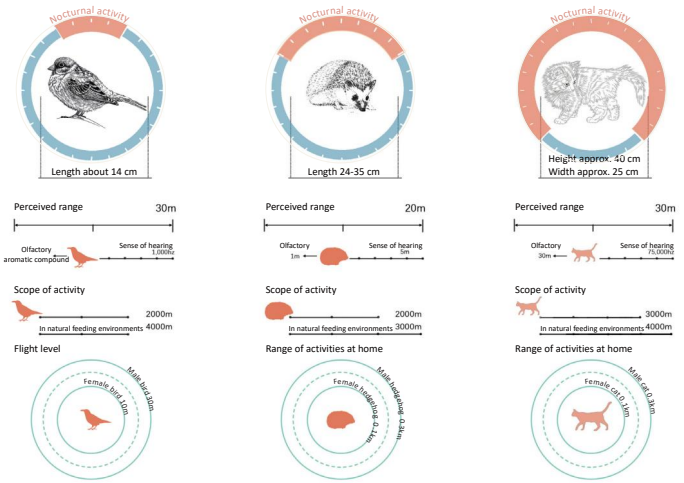
USER ANALYSIS

INVESTIGATE MAPPING





URBAN ANIMAL ANALYSIS



MASTER PLAN

Industrial Heritage Immersive District
1:Industrial History Museum
2:Industrial History Experience
3:Industrial Heritage Office Building

Recreation Industrial District
4:Recreation Centre
5:Nursing Home
6:Recreation and Nursing Industrial Park
7:Elderly University
8:Recreational Apartments
9:Community Activity Centres

Business district
10:Commercial Centre
11:Vibrant Market

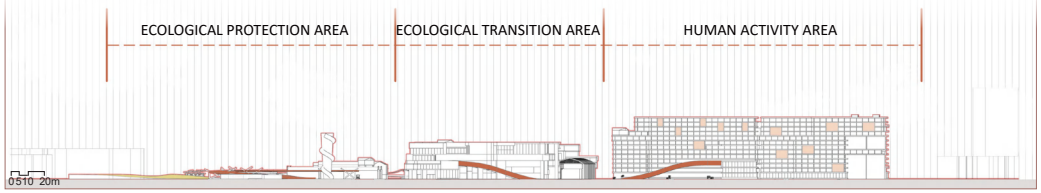
Interactive mode

Functional zone

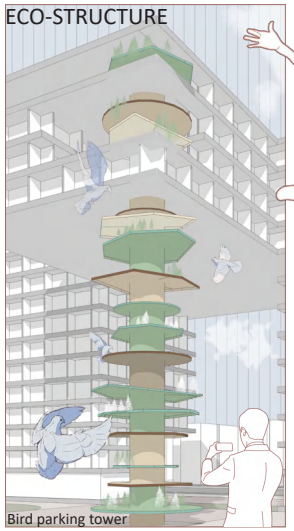
Traffic flow

1:Bird habitat
2:Transition zone
3:Stationary point
4:Interactive zone
5:Eco-promotional zone
6:Vibrant Market
7:Recreation Centret
8:Nursing Home
9:Commercial Centre
10:Recreation and Nursing Industrial Park
11:Elderly University
12:Recreational Apartments

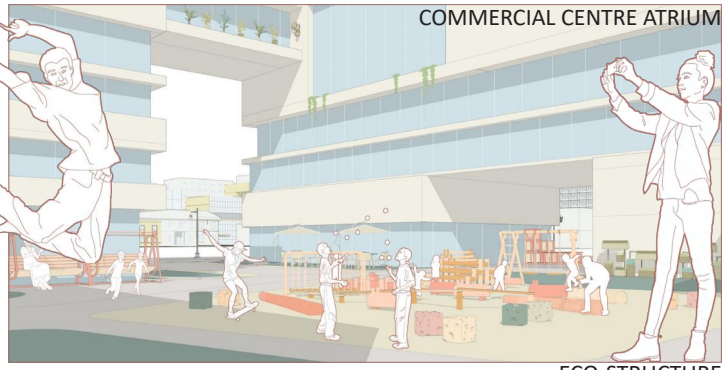
THE CITY SKYLINE



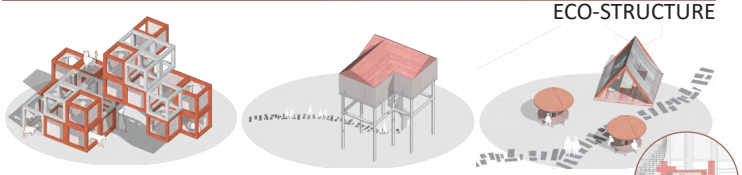
ECO-STRUCTURE

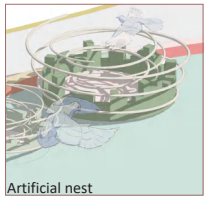


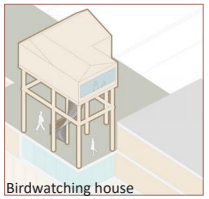
COMMERCIAL CENTRE ATRIUM

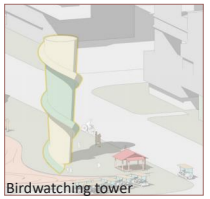


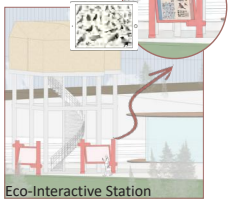
ECO-STRUCTURE



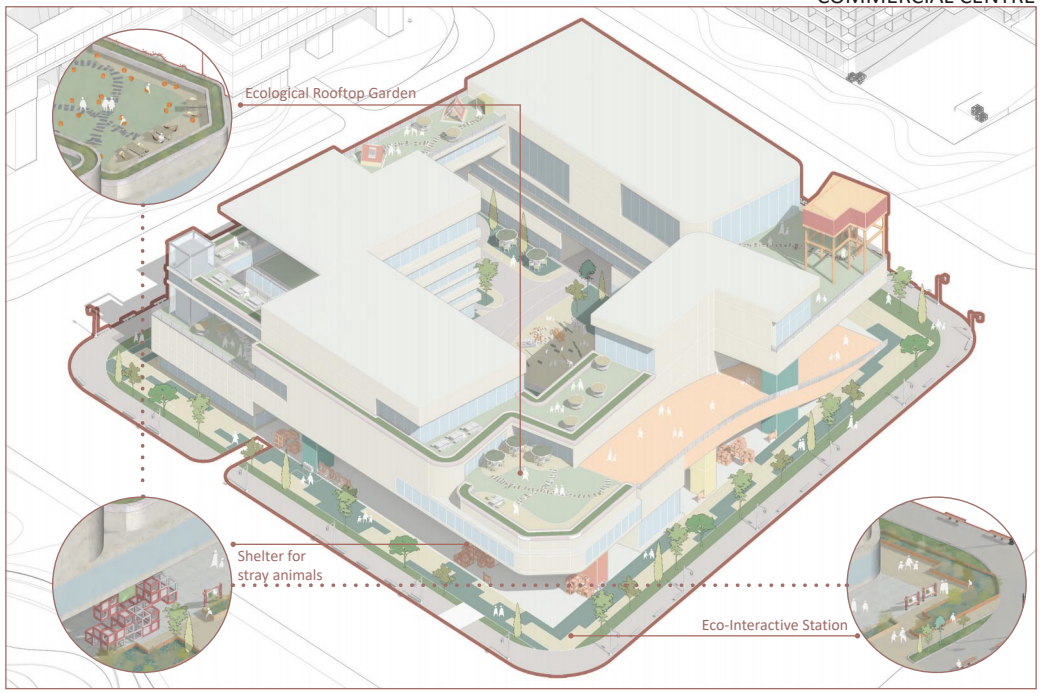








COMMERCIAL CENTRE

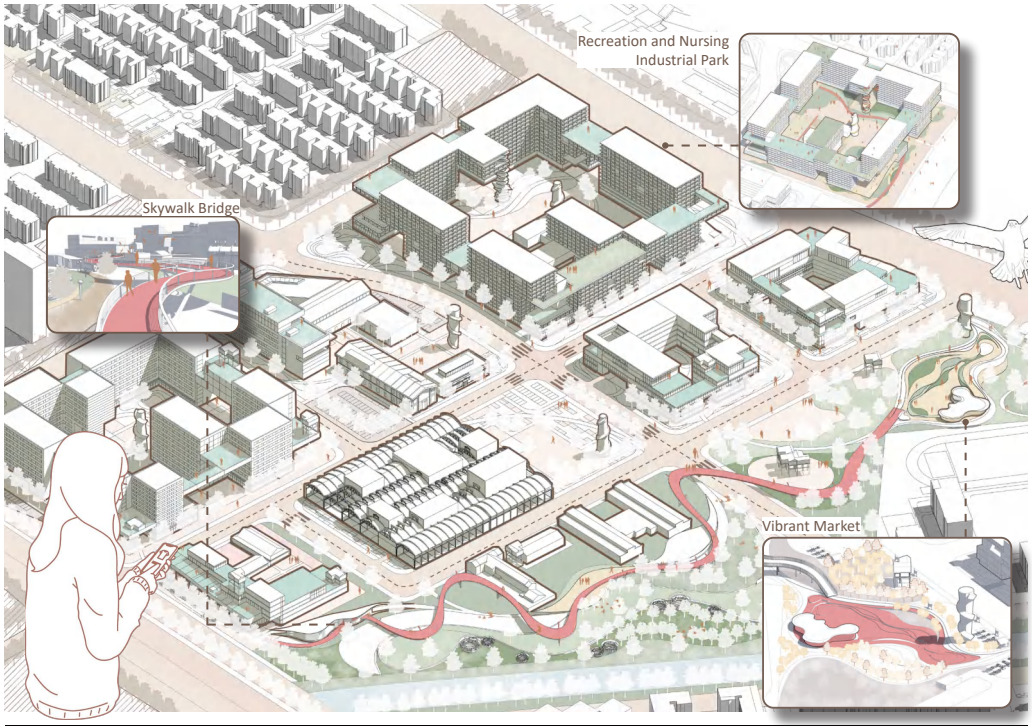


FACTORY CONVERSION

Qriginal workshop

Jin2 Workshop: Preserve portions of the structure and roof of the Jin2 Workshop for additions or alterations. The main functions of the Jin2 Workshop are currently a recreation centre, a library and a university for the elderly.

Qriginal workshop



MANUAL MODEL

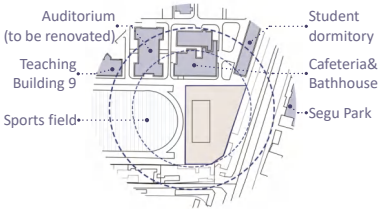
05

Sportzentrum

Design for Swimming Pool Cover Steel Structure Space Frame Steel Truss

Location:Hebei University of Technology. Tianjin,China
Area:4,900m² Academic Individual work
Duration:2024.6 Type:Architecture design

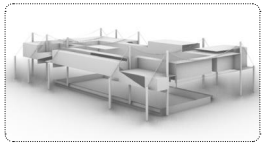
On the basis of utilizing and transforming the original grandstand and swimming pool, **SPORTZENTRUM** sets the ground floor as the functional space of the swimming pool, and the second floor as sports space & activity rooms.



STRUCTURE SELECTION

The structure needs to fulfill the requirements of a large span space on the ground floor and support a 2-story “return” volume.

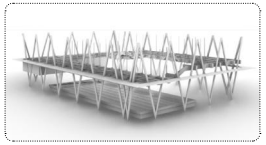
OPTION 1



SUSPENSION STRUCTURE

The first floor part is lifted up by the suspension cables on both sides.
Disadvantages: Suitable for large gymnasiums and other ultra-large-span buildings, the stability of the roof structure is not good, while the wind resistance is also insufficient.

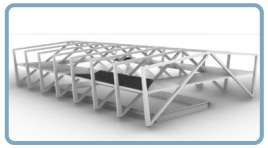
OPTION 2



V-SHAPED COLUMN +BOOM+TIC-TAC-TOE BEAM STRUCTURE

In which the inner side of the first floor slab is hung on the upper tic-tac-toe beam by the boom.Disadvantages: complex structure, high cost.

OPTION 3

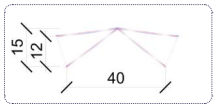


TRUSS CONSTRUCTION

All interior spaces are inside the trusses, which can address existing space usage needs.

Choose option 3 to continue the optimization, adjust the angle and height of the truss, and optimize the truss form by visualizing the result.

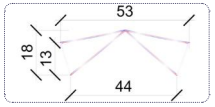
STRUCTURAL PROTOTYPE



Maximum structural utilisation of 0.57.
Maximum deformation is 0.636cm.
Structural weight of 153kg.

CONS: Too little structural utilisation.

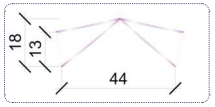
OPTIMISATION OPTION 1



The tilt angle of the outermost beam was adjusted by translating the articulation point, and it was found that the tilt angle had basically no effect on the structural utilisation and the maximum deformation amount, and only had an effect on the structural weight.

CONS:This approach has no effect on structural optimisation.

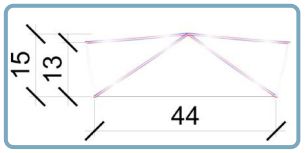
OPTIMISATION OPTION 2



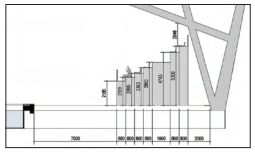
Adjustment of the upper suspension angle by adjusting the height of the uppermost articulation point.
Maximum structural utilisation of 0.82,Maximum deformation of 0.922cm,Structure weight 163kg.

CONS:Maximum stress exceeding the ultimate stress value of the material.

OPTIMISATION OPTION 3



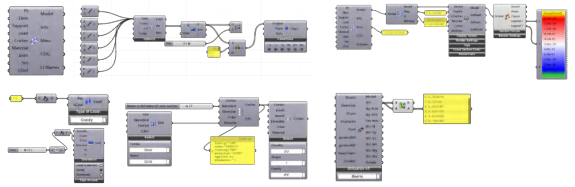
The utilisation of the structure is 0.03/0.04/0.5/0.5/0.6/0.7 respectively and the optimised utilisation increases.
The maximum stress in the structure is 2.1e+00kg/cm² which is within the material limit; The weight of the structure is 158kg and the steel usage has not changed much compared to the pre-optimisation.



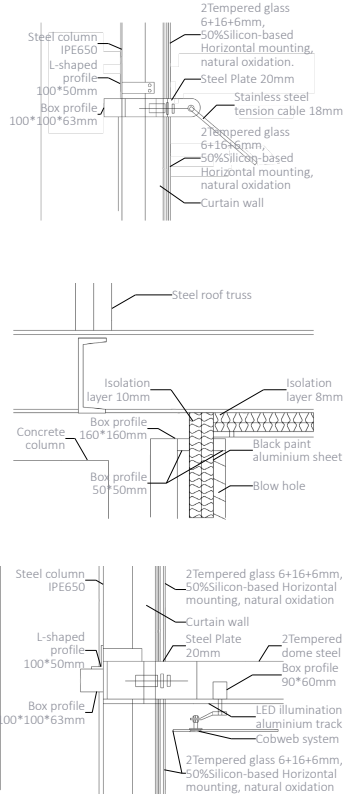
The human body sits at a viewing height of 1150mm, eyes to the top of the head of 120mm, the point of view to the first row of 7500mm. take an arbitrary order calculation method, publicised as follows:

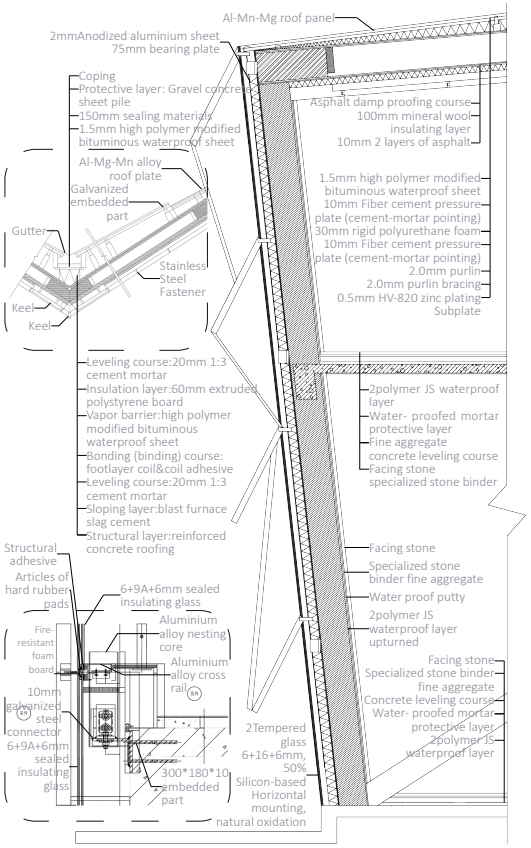
$$Y'_n=Y'_n(\tan\alpha'+2.3026\ \frac{C}{d}\ \log\frac{X_n-0.5d}{X_1-0.5d})$$

$$\tan\alpha'=\frac{Y'_1}{X'_1}$$

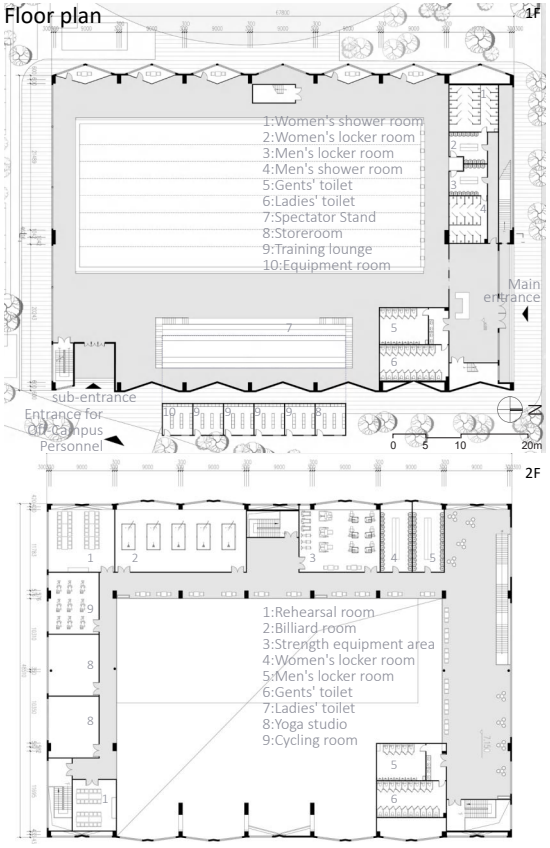


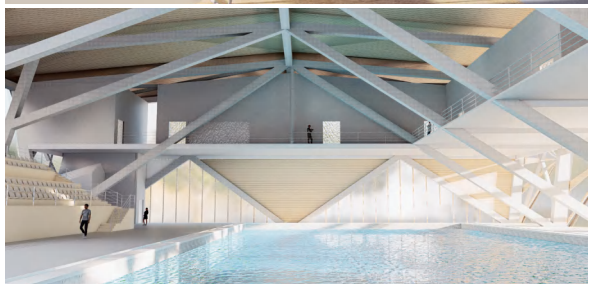
Architectural detail drawing

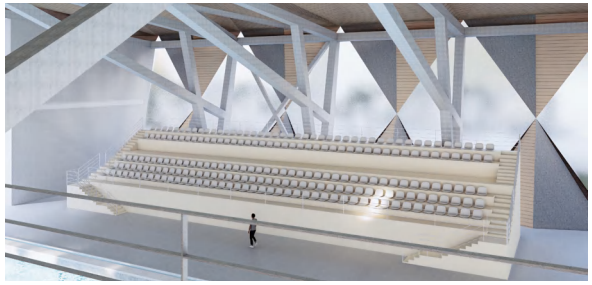




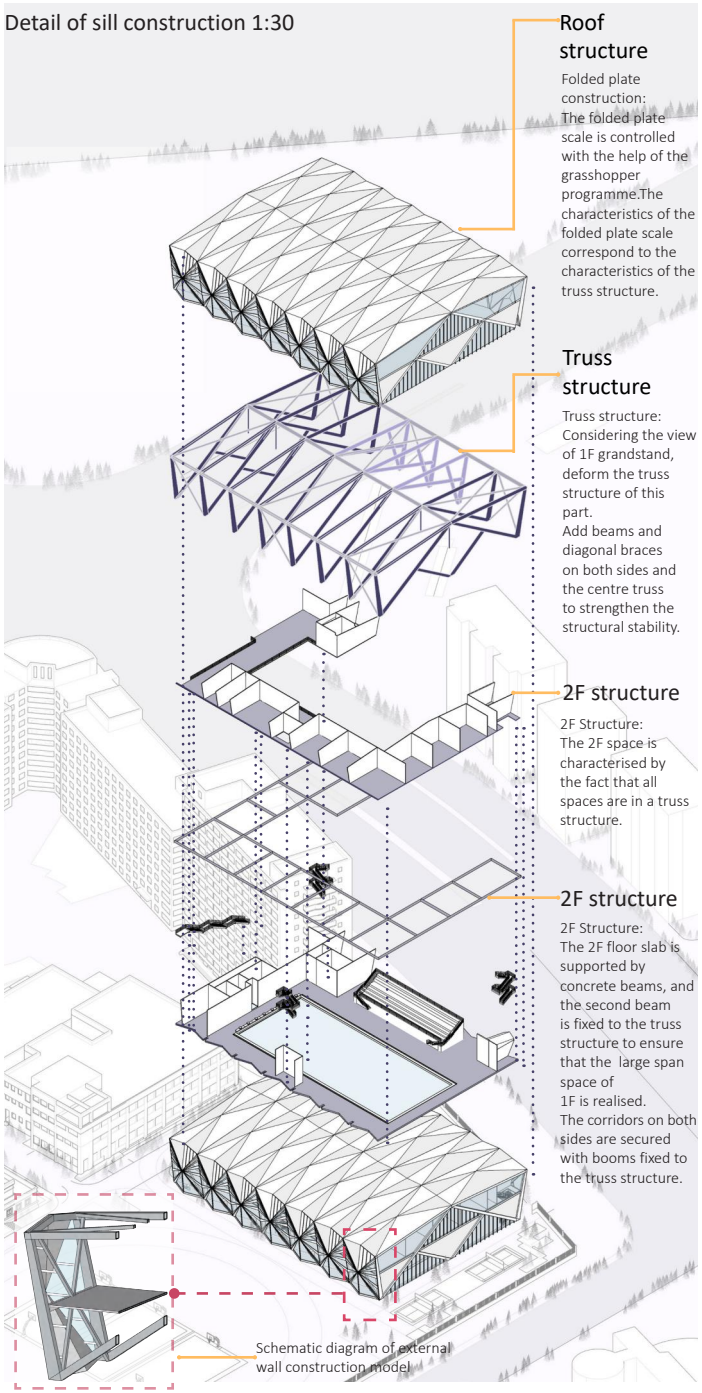
Floor plan



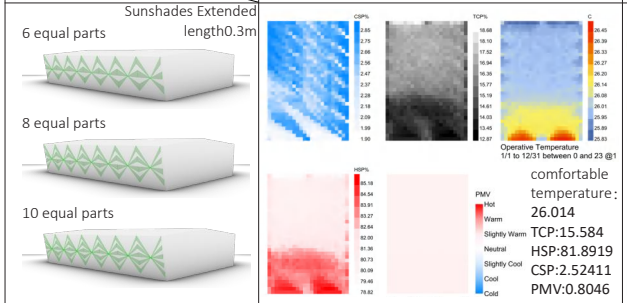
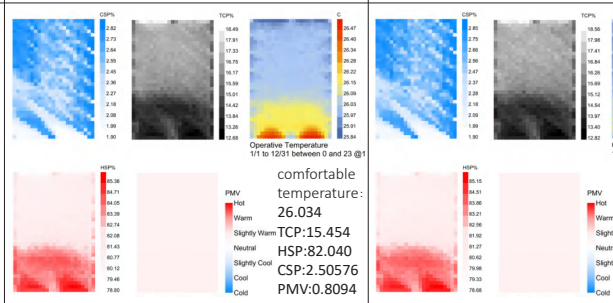
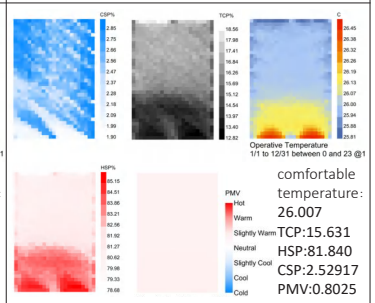
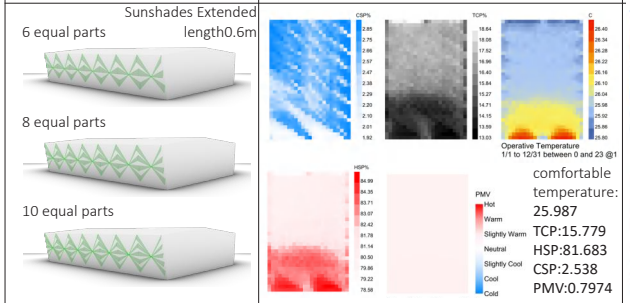
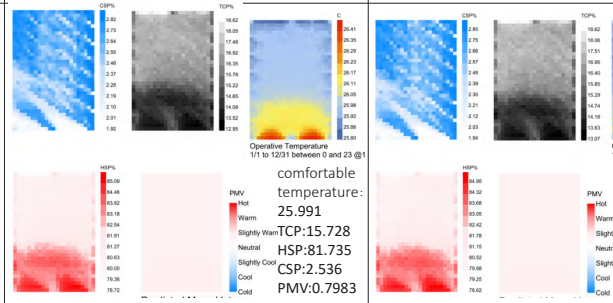
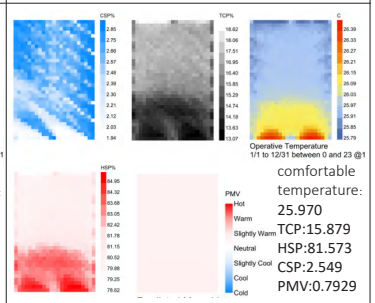
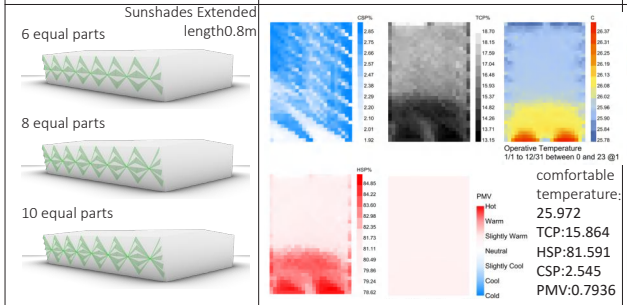
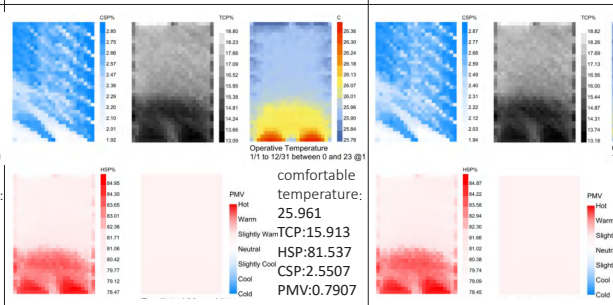
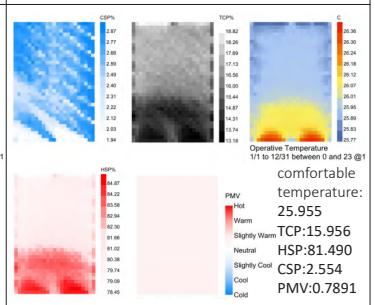
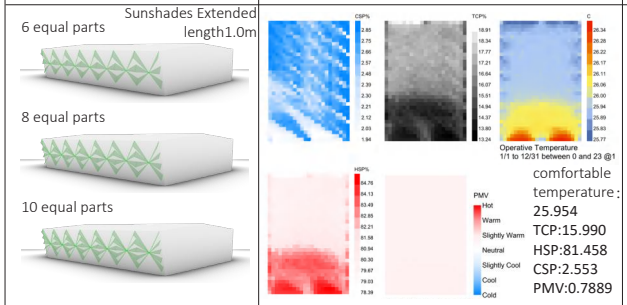
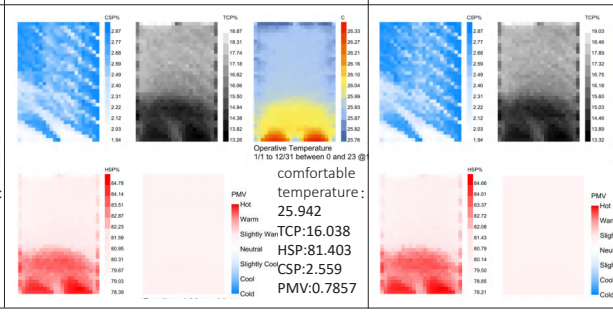
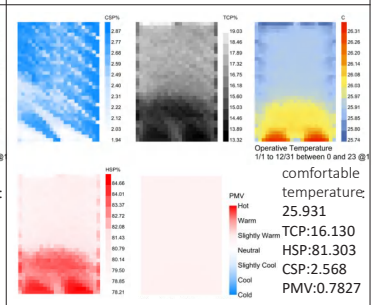




Detail of sill construction 1:30



Visualisation of epidermal optimisation processes and indices

Model \ Number of sunshade	3 consecutive sunshades per set Extended length0.3/0.6/0.8/1m	8 consecutive sunshades per set Extended length0.3/0.6/0.8/1m	10 consecutive sunshades per set Extended length0.3/0.6/0.8/1m
Sunshades Extended length0.3m 6 equal parts 8 equal parts 10 equal parts			
Sunshades Extended length0.6m 6 equal parts 8 equal parts 10 equal parts			
Sunshades Extended length0.8m 6 equal parts 8 equal parts 10 equal parts			
Sunshades Extended length1.0m 6 equal parts 8 equal parts 10 equal parts			

Optimisation result analysis

Vertical	The longer the sunshade extends, the greater the proportion of thermal comfort area& the most comfortable at 1M extends.
Horizontal	Comparison of the TCP visualisation charts shows that the number of sunshades is most comfortable when the number of sunshades is 4.

The main indoor activity space is most comfortable when the number of shades is 4 and the extension length is 1M.